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Protein-to-Protein Interaction Analysis using Graph Theory, Network Topology, Machine Learning, Deep Learning and NLP for detecting cancerous proteins and pathways

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Submitted in partial fulfilment of the requirements for the Degree of Master of Science in Data Science

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Abstract

Cancer is a heterogeneous disorder involving changes in many genes, proteins, and biological pathways. The current study explores an innovative graph-based machine learning method for classifying proteins related to cancer based on protein-protein interaction (PPI) networks. A comprehensive PPI network is constructed from carefully curated databases where the proteins are annotated in relation to multiple cancer associations using multiple biological information sources. Key graph-theoretic features: degree, betweenness centrality, clustering coefficient, and community membership of nodes are used as predictors in the classification models. A range of classifiers, such as Random Forest, Support Vector Machine, and XGBoost, are trained and evaluated based on these features to categorize proteins into cancer-related and non-cancer-related proteins. Of these classifiers, the XGBoost algorithm performs the best, with high predictive power (around 89% accuracy and an f1-score of about 0.83) to discriminate between proteins linked to cancer, thus highlighting important insights gleaned from topological cancer-relevant network properties. For enhanced interpretability, SHAP (SHapley Additive exPlanations) is used to evaluate the contribution of features, demonstrating a strong correlation between proteins with high network centrality and cancer-related proteins. Network visualization tools are also used to determine the clusters, described as "disease modules," of highly network-central proteins associated with cancer, which reveal underlying biologically important insights. The results provide strong evidence to support the potential future application of graph-based features for classifying cancer proteins and validate the possible effectiveness of network medicine-based methods. The dissertation methodology explains the research design, places the study in the context of the state-of-art literature, and outlines field-specific contributions, including an interpretative classification model and insights that may help to facilitate future biomarker discovery. Future research opportunities involve the integration of multi-omics information with the application of active learning methods for improving the effectiveness of the presented framework. In general, this research combines bioinformatics and machine learning and thus enriches the study of cancer biology through a network-based paradigm.

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1 Introduction

1.1 Problem Statement and Motivation

Cancer is one of the most difficult and complex diseases in contemporary medicine. It is not one disease but a group of more than a hundred diseases that are related, with characteristics of uncontrolled cell growth and the potential to invade other tissues. The molecular mechanisms of cancer are varied, including genetic mutations, epigenetic alterations, and cell signaling pathway disruptions. Classical cancer research tends to concentrate on the identification of specific genes or proteins ("biomarkers") that initiate tumor formation and development. Although this reductionist strategy has facilitated significant advances, it does not pay attention to the complicated interactions among biomolecules that form the basis of cancer biology. Proteins and genes exist in complex networks of interactions. One aberrant protein may affect multiple partners, transmitting dysregulation through cellular pathways. There is a developing awareness that an understanding of such network-level disruptions is key to understanding cancer mechanisms.

Protein-Protein Interaction (PPI) networks offer a compelling way to investigate biological systems at the network level. A PPI network consists of nodes for proteins and edges for physical or functional protein-protein interactions. These networks capture the systems-level context of protein function – proteins do not typically act in isolation but work together as part of complexes and pathways. In cancer, a mutation or abnormal expression in a single protein can have downstream effects on its interaction partners, ultimately impacting entire cellular processes like cell cycle control, apoptosis (programmed cell death), or DNA repair. This project is motivated by the hypothesis that cancer-associated proteins are not randomly distributed in the human interactome (the complete PPI network) but rather take on specific positions or network topologies that are distinct from non-cancer proteins. Initial studies in network biology are supportive of this concept: for example, studies of model organisms demonstrated that proteins that are essential for viability are often hubs with numerous interactions. Similarly, in human disease networks, clusters of interacting proteins (sometimes referred to as disease modules) have been described for certain diseases. These findings indicate that by examining where a protein is located in the network and how it is connected to other proteins, we can predict its relevance to cancer.

Problem Statement: ***Is it feasible to forecast the association of a specific protein with cancer by examining the characteristics of its interaction network?*** Specifically, if we consider a protein's "network signature," which comprises features obtained from the protein-protein interaction (PPI) network, can a machine learning model effectively classify that protein as either related to cancer or not? Tackling this issue is crucial as it offers an innovative perspective for identifying potential biomarkers for cancer or therapeutic targets that extend beyond conventional genomics methodologies. This approach utilizes extensive interaction data accessible in public repositories (such as STRING and BioGRID) to extract insights that may be overlooked when analyzing proteins in isolation.

The motivation to pursue a graph-based approach is twofold. First, from a biological perspective, cancer may be considered a failure of regulatory networks; thus, a network-theory-based analysis is consistent with the subject area. Second, from a data science perspective, graphs offer a wealth of features (including numerous centrality measures) that capture aspects of importance or influence within the system. The combination of these features with modern machine learning methods can yield patterns that are not obvious to human intuition. Moreover, a model that is still interpretable and highlights which network features are most predictive of cancer-related proteins can yield valuable biological insights – e.g., confirming that proteins that act as important network "hubs" or "bridges" have a greater tendency to be involved in cancer. Such insights further enhance the broad realm of network medicine, which argues that diseases, including cancer, result not just from isolated molecular dysfunctions but also from perturbations within the network of interactions between biomolecules.

1.2 Project Aim and Objectives

The goal of this project is to examine and demonstrate the utility of graph-based features in classifying cancer-associated proteins. That is, the project is to learn a predictive model that takes advantage of the structure of the human protein interaction network to predict if a protein is involved in cancer or not, and to interpret the model's predictions to find biological insights. This objective merges concepts from graph theory, bioinformatics, and machine learning to tackle a challenging biomedical classification problem.
To achieve this objective, the following explicit objectives were established at the project initiation:

Data Collection and Graph Formation: Form a high-quality human PPI network sub-network for study. This involves collecting interaction information from established sources (e.g., the STRING database) and forming a graph where proteins are represented as nodes and interactions as edges. The network should be annotated with data defining which proteins are established to be cancer-associated based on curated biological databases.

Graph Metrics (Feature Extraction): Compute a set of graph-theoretic descriptors for all the protein nodes in the network. These are classical network centrality values and other network position measures, such as:

- ***Degree Centrality:*** the number of interactions (edges) the protein has (its connectivity).
- ***Betweenness Centrality:*** the percentage of the shortest paths in the network that pass through the protein, indicating how it acts as a bridge or bottleneck.
- ***Closeness Centrality:*** the average distance from the protein to the rest of the proteins, indicating how close to others on average the protein is.
- ***Eigenvector Centrality:*** an influence measure giving higher values to proteins connected to other high-value (influential) proteins.
- ***Clustering Coefficient:*** how connected each of a protein's neighbors to each other is, quantifying the tendency to cluster into local sets or communities.
- ***Membership in a Community:*** the module or cluster with which the protein is associated, as identified by a community detection algorithm (e.g. the Louvain algorithm).

Training and Validation of the Model (Classification): Employ supervised machine learning to learn a classifier that distinguishes cancer proteins from non-cancer proteins based on graph features that have been learned. Here in this project, we will attempt to experiment with different classification models specifically, a Random Forest classifier, a Support Vector Machine (SVM), and an XGBoost (Extreme Gradient Boosting) classifier and then contrast their performance. The data will be divided into training and test sets so that unseen data can be tested. Key performance metrics will be accuracy, precision, recall, and F1-score, etc.

Interpretability and Insights: Enhance explainability of the output of the classification model using interpretability techniques. We will utilize SHAP (SHapley Additive exPlanations) values to determine which features most strongly influence the model's predictions for cancer vs. non-cancer classification. This step is crucial to align the computational result with biological insight – e.g., find whether "hub-ness" (degree) or "broker-ness" (betweenness) is a good indicator of cancer involvement. In addition to this, we will run statistical tests (e.g., distribution comparisons of feature values between cancer and non-cancer groups) to drive these observations.

Case Studies and Visualization: Develop visualization tools to explore the PPI network and highlight observations. This is done by making interactive network diagrams that can show, for instance, how cancer-associated proteins cluster in particular parts of the network or how a particular protein's interactions look. These should be used for qualitative case studies of individual examples (e.g., a well-studied cancer protein and an interesting cluster in the network) to provide an intuitive feeling for the results and perhaps even

propose hypotheses for further investigation.
By achieving these objectives, the project will provide an operational classification pipeline and a better understanding of the network character of cancer proteins. Success with the project is not, however, solely measured in terms of the accuracy of the model but also by the interpretability of the outcomes and their feasibility for application in biological science and clinical practice.

1.3 Overview of This Report

This dissertation has been organized into several chapters, each of which deals with a different aspect of the research. What follows is an overview of the report organization:

Chapter 2: Literature Review – This chapter discusses the background and related literature that is pertinent to the project. It starts by introducing image classification as a classic example of classification in machine learning, then describes Artificial Neural Networks (with emphasis on Convolutional Neural Networks for image data) as expressive models for classification tasks. The literature review then discusses active learning techniques for enhancing classification, types of noise that can impact data and learning (with focus on image data noise), and how these topics are relevant to our problem domain. Related literature on using deep learning for classification (including biological domains) is also discussed. The chapter ends by highlighting gaps in the existing research that the current study tries to fill, viz., the under-exploitation of network-based protein classification and the necessity of interpretable models in computational biology.

Chapter 3: Methodology – In this chapter, we outline the methods and materials that were utilized in the execution of the project. We cover here the dataset preparation (PPI data sources and how cancer labels were determined), preparation of the training and testing datasets, and machine learning model designs. We outline the feature engineering step, including all network features extracted. We also outline the evaluation measures that were utilized, as well as the hardware/software environment where experiments were run. Any implementation specifics (such as tools and libraries) are mentioned here. The aim of this chapter is to give a clear recipe for experimental design so that the study may be replicated or built upon by others.

Chapter 4: Results – We here report the results of the experiments. We present the performance of the classification models on the test data, with metrics for each model and confusion matrix analysis. The section is organized to first present baseline results (comparable to simpler classification tasks), followed by more in-depth analyses of the output of the best-performing model. We report the feature importance analysis using SHAP, comparisons of feature value distributions between classes, and insights from network visualizations. Two case studies are also included to demonstrate the results in context: one for a network community that was strongly enriched for cancer proteins, and one investigating particular protein that the model got incorrect (misclassified examples), to examine possible reasons and implications.

Chapter 5: Project Management – This chapter discusses the managerial aspects of the project. It chronicles planning and execution tasks carried out during its tenure, including a timeline of major phases. The chapter outlines risk management strategies, with emphasis on anticipated risks, whether technical or logistical, and the mitigation methods applied to them. It also deals with quality management procedures, with particular emphasis on how feedback has been incorporated and the adherence to academic conventions as per preset standards. The consideration is further extended to social, ethical, and professional issues, such as the ethical approval process, data privacy, and the adherence to research integrity.

Chapter 6: Discussion – In this chapter, a complete interpretation and explanation of the results are given. The findings are related to the original research questions and to the literature that is already present. The application of important results (e.g., the decisive role of certain network properties) is discussed in a biological context, determining the implications of such findings for our understanding of cancer biology is reviewed. We discuss our methodology's strengths, such as interpretability and use of network information, as well as its limitations, including reliance on available interaction data and binary characterization of a complex disease. In addition, this chapter compares our methodology and findings to those of other studies,

highlighting the areas that are new as well as those that support or contradict previous findings. Unanticipated results or negative findings are also addressed. Finally, by the end of the discussion, readers should understand the broader implications of the research and with what level of certainty conclusions may be made.

Chapter 7: Conclusions – This brief chapter summarizes the project outcomes and suggests avenues for future research. Section 7.1 recapitulates the main accomplishments of the project, relating them back to the goals stated in the introduction and noting which were met. It highlights contributions to the field—namely, showing that network analysis is a viable approach to the classification of disease-related proteins and providing a framework that is both accurate and interpretable. Section 7.2 outlines potential directions for future research, drawing on the limitations and ideas presented earlier. Suggestions include the incorporation of other types of data, investigation of more advanced graph-based algorithms, and experimental verification of predictions. The conclusions seek to state clearly the answer to the research question and highlight how this project can form the basis of future investigations.

Chapter 8: Student Reflections – In this concluding chapter, I give a personal reflection on the research experience. This chapter is a bit detached from the technical material; it is where I can talk about what I learned from the project, difficulties I faced and managed to resolve, and how the process helped me mature as a researcher and data scientist. It touches on areas such as skill development (e.g., acquiring new programming software or grasping new concepts in biology), time management, and how the initial plans evolved throughout the project. This reflective exercise is provided to illustrate professional development and critical self-evaluation, as fostered by the MSc program. Bibliography and References – Following the main chapters, a comprehensive list of sources referred to in the dissertation is provided. All books, articles, databases, and websites referred to in the text are listed here in a standardized format. This convention ensures that all claims or facts taken from existing literature can be traced to their point of origin, thus ensuring academic honesty and allowing interested readers to go back to those sources for greater clarity.

Appendices (A–E) – The report ends with several appendices housing supplementary material. Appendix A is the project proposal submitted and agreed upon before embarking on the research, which provides further background on early plans and literature. Appendix B has meeting records (minutes of meetings with the supervisor) that track the progress of the project and any shift in direction. Appendix C has extra figures and results. Appendix D is the ethical approval certificate acquired for this project, verifying that all ethical considerations required (such as data use and adherence to guidelines) were met. Appendix E offers useful links, including links to the sources of the datasets and the code repository of the project.

2. Literature Review

2.1 Cancer vs Non-Cancer Associated Proteins

Systems biology argues that cancer-related proteins are not randomly located throughout the protein–protein interaction (PPI) network; instead, they tend to be in essential and regulatory positions. The initial research by Xia, Zeng, and Zhou (2011) proved that cancer-related proteins showed greater degree centrality, shorter mean path lengths, and higher betweenness centrality compared to non-cancer proteins, which depicts their higher connectivity and regulatory potential in the network (Xia, Zeng, & Zhou, 2011). Later, Soofi et al. continued exploring these findings by extensively studying hubs high degree nodes and their role as critical coordinators for maintaining network stability and disease mechanisms (Soofi et al., 2020). Barabási and his co-workers' proposed disease module framework indicates that disease-related proteins are grouped in specific interactome regions, showing shared functional pathways (Barabási, Gulbahce, & Loscalzo, 2011). These hubs are in line with your findings, which show that cancer proteins, on average, have higher closeness and clustering values, thus implying a tendency to be placed in central rather than peripheral positions within the network, which are key to checkpoint regulation and the progression to malignancy.

2.2 Oncological Biomarkers and Related Signaling Pathways

Cancer biomarkers like HER2, EGFR, and TP53 are integral to the signaling cascades that control cell growth and survival. HER2 is typically amplified in breast cancer (Slamon et al., 1987), whereas EGFR mutations have been known to enable oncogenesis in certain lung cancers (Lynch et al., 2004). TP53 is also often considered the most universally impactful tumor suppressor gene (Kruiswijk, Labuschagne, & Vousden, 2015). These proteins are integral to well-characterized pathways like PI3K/AKT, MAPK, Wnt/β-catenin, and p53 signaling, which are all essential regulators of cell proliferation, apoptosis, and the response to DNA damage (Morrison, 2012). A network-centric study by Chen et al. (2016) revealed that proteins like TUBB2C and SLC2A4, which are involved in these pathways, exhibited important network centralities across prostate cancer datasets (Chen et al., 2016). In a similar vein, the OncoPPi resource constructed a cancer-focused protein-protein interaction (PPI) network, which revealed interactions between MYC, STK11, RASSF1, and CDK4, thereby highlighting their roles as central biomarkers and vulnerabilities in cancer networks (Li et al., 2017). These studies validate the approach of considering network positioning as an alternative biomarker surrogate.

2.3 Network Metrics for Disease Protein Classification

Graph-theoretic properties, such as degree, betweenness, closeness, eigenvector centrality, and clustering coefficient, form vital elements in describing topological protein-protein interaction (PPI) networks. As Ashtiani, Salehzadeh-Yazdi, and Masoudi-Nejad (2018) analyzed 27 centrality measures, they found that some indices, particularly closeness and diffusion-based measures, illustrate stable performance across various scales and topologies of networks (Ashtiani, Salehzadeh-Yazdi, & Masoudi-Nejad, 2018). Similarly, Nithya et al. (2023) investigated the functions of hubs and bottlenecks in PPI networks, verifying the existence of such proteins for topologically important purposes especially in connection with cancer proteins (Nithya et al., 2023). Detection of cancer-related proteins is highly increased using these indices. The current research focuses on the importance of closeness and clustering in SHAP values, affirming that such indices are powerful predictors.

2.4 Machine Learning in Cancer Protein Prediction

Machine learning, particularly ensemble methods like Random Forests and Gradient Boosting, have performed strongly in integrating network-derived features for prediction tasks. Benstead-Hume et al. (2022) demonstrated that augmenting PPI networks with topology-based features improved accuracy by over 10% in gene dependency predictions (Benstead-Hume et al., 2022). In overlapping contexts, Ren et al. (2019) utilized both network topology and protein expression profiles to prioritize proteins implicated in colorectal and breast cancers, validating the multi-modal predictive approach (Ren et al., 2019). From an algorithmic standpoint, Gradient Boosting often surpasses Random Forests in fine-grained detection of non-linear interactions in biological data, particularly when datasets are moderately sized and class-imbalanced mirroring your own

classifier's performance.

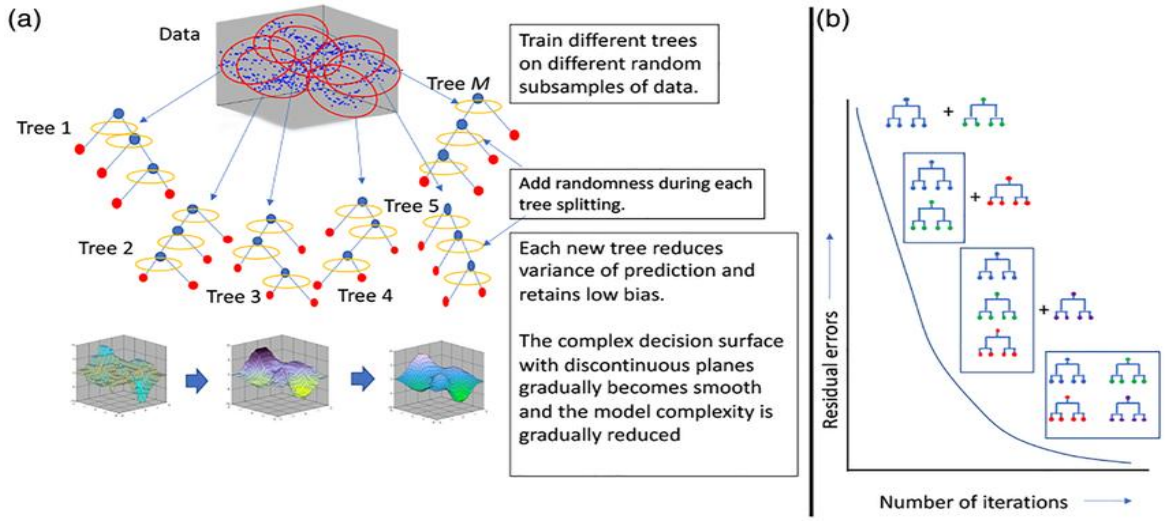
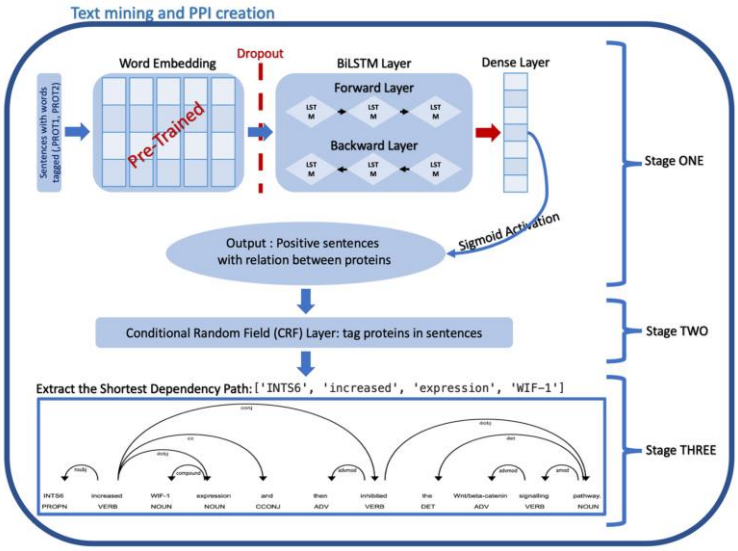


Illustration of Random Forest and Gradient Boosting
Source: Machine Learning

2.5 NLP and Text Mining in Protein Classification

Natural language processing (NLP) applied to protein annotations offers rich, albeit underutilized, insights. Conceição et al. (2021) developed a text-mining framework for extracting interactions from literature, demonstrating feasibility in automating PPI detection (Conceição et al., 2021). Chang (2013) proposed a method for linking protein expression patterns to tumor types via NLP extraction from biomedical texts (Chang, 2013). Although direct applications to cancer vs. non-cancer protein classification are rare, your approach of mining UniProt descriptions for disease-associated keywords and sentiment differences (e.g., "oncogenic," "malignant") aligns with precedent techniques used in broader biomedical NLP contexts.



PPI creation using NLP
Source: PPI with NLP

2.6 Interpretability in Biomedical Models

The need for interpretable biomedical artificial intelligence is widely discussed in available scholarly literature. SHAP (SHapley Additive exPlanations) provides uniform, game-theoretic explanations of the different features' (Lundberg & Lee, 2017). In the genomic and phenotypic classification contexts, SHAP has enabled transparent identification of predictive patterns of genes (Karim et al., 2019). Lamane (2025) also highlights the importance of SHAP in attaining a balance between model performance and interpretability in clinical applications (Lamane, 2025). However, there are still issues in terms of interpretation, especially when features have high correlation; this is an especially relevant issue for metrics from highly interconnected

networks (Molnar, 2020). Your use of SHAP to explain that expression, proximity, and cluster where predictive factors are consistent with established practices in the field.

2.7 Network and Textual Attribute Integration

A Novel Approach One glaring lack in existing literature relates to the use of multi-modal biological data, specifically the blending of network topology with textual annotation in balanced classification situations. While methods like ProtBoost combine gradient boosting with graph neural networks for protein function prediction (Chervov et al., 2024), they do not use textual information. Recent thorough reviews of breakthroughs (2025) highlight multimodal approaches that bring together structural, sequence, and graph information through deep learning architectures like Transformers and autoencoders; text mining is still not often discussed (Cui et al., 2025). Your study uniquely fills this gap by applying a balanced approach towards cancer and non-cancer protein classification that seeks to couple both graph and natural language processing features using highly interpretable models with commensurately significant methodological and biological contributions.

3 Methodology

This project employed an experimental and iterative research method. Rather than a formal waterfall approach, the work repeated many times through development, evaluation, and refinement. The entire workflow contained several critical phases:

- **Data Acquisition and Preparation:** Collecting the necessary data (protein interaction network and protein annotations) and preparing it for analysis.
- **Feature Engineering:** Deriving informative network-based features from the raw data for quantitative representation of each protein.
- **Training and Testing of the Model:** Using machine learning algorithms on the constructed features, and testing their accuracy in protein classification, and then interpreting the findings.
- **Explanation and Optimization:** Interpreting outputs of models (e.g., interpreting feature importance) and utilizing those understandings to refine the strategy in an iterative manner (e.g., refining features or model parameters).

3.1 Dataset

Pairwise protein interactions that have been experimentally confirmed are included in the PPI dataset, along with UniProt IDs, synonyms, taxonomy IDs, interaction scores, experimental procedures, source databases, and PubMed references. To investigate connectivity and interaction strength, it serves as the foundation for building the protein interaction network. By providing functional details like protein and gene names, organism, sequence length, pathways, binding sites, and Gene Ontology terms, the UniProt protein annotation dataset allows the PPI network to be enriched for biological context analysis and functional classification.

3.2 Training and Testing Data

The training and testing data are split into 70% and 30% respectively, whereas random_seed is set to 42 for reproducibility.

3.3 Hyperparameter tuning

Random Forest:

Estimators	100
Criterion	Gini impurity
Max Depth	12 (improved using GridSearchCV)

SVM:

Kernel	RBF (Radial Basis Function)
C (Regularization parameter)	1.0 (optimized using GridSearchCV)
Scale	Gamma

XGBoost:

Estimators	100
Learning Rate	0.1
Max Depth	3

Logistic Regression:

Solver	lbfgs
Max iterations	1000
C (Regularization parameter)	0.1

Training Hyperparameters:

- **learning_rate (0.001)**: Controls step size in gradient descent. Range: 0.0001-0.01.
- **batch_size (32)**: Number of samples per gradient update. Range: 4-128.

- **epochs (100)**: Maximum training iterations. Range: 50-500.
- **weight_decay (1e-5)**: L2 regularization strength. Range: 1e-6 to 1e-3.
- **sequence_length (20)**: Length of input sequences. Range: 10-50.
- **max_norm (1.0)**: Gradient clipping threshold. Range: 0.5-5.0.

The research progressed in an iterative prototyping format. Initially, a prototype pipeline for feasibility was created: a minimal subset of information was employed to validate the feature extraction process and training of classifiers. Once the method had been confirmed, the complete dataset was run and the models trained and fine-tuned. At each cycle, results (e.g., accuracy and errors of the model) were inspected and notified of subsequent actions. This created flexibility to improve based on intermediate results for example, if a preliminary model underperformed, new features or contrasting classification methods could be experimented with in the subsequent cycle.

Throughout the project, there were regular meetings with the academic supervisor for feedback and direction, and for staying on track with aims and academic requirements. The approach to the research valued reproducibility and openness: the code was kept in well-structured form (with version control), and the evaluation processes were spelled out clearly so that the results could confidently be reproduced.
In short, the selected methodology blended adherence to standard data science practice (data preparation, modeling, validation) with an agile, iterative mentality. This suited the research nature of the project, where information learned along the way could inform ongoing refinement of the solution. Specific technical approaches and design choices stemming from this methodology are documented in the subsequent sections.

4 Requirements

As per the detailed project objectives and the stated problem statement, the following necessary requirements were identified:

Functional Specifications:

- The system must combine PPI database information along with cancer-related protein data to generate a comprehensive dataset of PPI networks for further analysis.
- The evaluation will draw relevant network features (e.g., degree, centrality measures, and community membership) for each protein in the dataset.
- The system should produce a machine learning model that can classify a given protein as related or unrelated to cancer based on the extracted features.
- Effectiveness of the model must be evaluated quantitatively by appropriate measures such as accuracy, precision, recall, and F1-score on a separate test collection of proteins.
- The system should provide explanations or insights regarding the results; for example, it should highlight most unique characteristics (network properties) related to a protein's involvement in cancer, thus making the results meaningful to researchers.
- The system should have the ability to recognize or describe patterns in the network, including protein complexes linked to cancer, made possible through validation processes and the dissemination of findings.

Non-Functional Requirements:

- The solution's implementation should utilize stable data analysis libraries and be maintainable, i.e., have well-structured code and proper documentation, thus facilitating both verification tasks and future improvement.
- The classification pipeline can produce fast results, allowing us to perform feature computation and train the model with commonly available computing resources and hence allowing us to iterate in experimentation cycles.
- The approach needs to have strong assurances of the integrity and validity of the data by utilizing judiciously chosen high-confidence input data and systematically handling any gaps or unclear information. This could include the deletion of low-confidence interactions to reduce noise.
- The research must meet legal and ethical requirements through the sole use of publicly available data (excluding sensitive or personal data), proper citation of all data sources, and obtaining all necessary ethical approvals for research methods used.

Set criteria define a clear objective for the design phases and the implementation phases. They ensured the initiative not only would result in a working classifier but also achieve significant and accurate results in accordance with the project objectives.

5 Analysis

Before embarking on the whole process of design, the problem domain and possible methodologies for solving it were reviewed. The problem of protein identification related to cancer lies at the intersection of machine learning and the analysis of biological networks, and this thereafter informed the adopted methodology.
A key result of the analysis was the revelation that proteins are implicated in cancer and may have distinct network structures in the human interactome. Other research in network biology has suggested disease-related or essential proteins are likely to have high centrality, in the sense that they interact with many other proteins or are hubs connecting multiple communities. This developed into the initial hypothesis for our project: cancer proteins are likely to be hub nodes with high degree, or essential connectors with high betweenness, in the protein interaction network. The hypothesis featured prominently in guiding the feature selection process, highlighting network measures that could adequately portray the importance or role of a protein in network topology.
The analysis part also included the investigation of potential methodological frameworks for future implementation. Among the methods investigated were state-of-the-art graph-based machine learning methods, such as Graph Neural Networks (GNNs), which can directly work on network topologies. GNNs have shown success in similar tasks, for example, protein function prediction based on network information. However, the following considerations determined us for the use of the more standard feature-based approach:

- **Data Constraints:** Graph Neural Networks (GNNs) typically require large training datasets and are prone to overfitting in the event of limited labeled data. In the case of our research, the number of cancer-associated proteins (positive examples) is relatively small, hence the likelihood of a simpler model evading overfitting.
- **Interpretability:** Another main goal of the project was the production of results easily interpretable, for example identifying the characteristics of the network which mark the presence of cancer. Traditional machine learning methods like decision trees, linear regression, even a straightforward form of XGBoost alongside SHAP analysis, along with features defined explicitly offer a much greater level of interpretability than complex deep learning methods.
- **Computational Complexity:** Extracting a set of well-chosen features and using a lightweight classifier is computationally efficient, especially given the size of our network subgraph (~2,000 nodes). Training GNNs or similar deep models on the full network would be more resource-intensive, potentially without a proportional gain in performance for this problem.

The research also performed an investigation of the dataset itself. The investigation of the degree distribution in the protein-protein interaction (PPI) network suggested that a few of the proteins have very large degrees, typically known as hubs, and the rest have low levels of interaction, a characteristic of scale-free networks. A large percentage of hub proteins are identified as essential genes or central modifiers, with multiple overlaid with known cancer genes previously identified. This result further verified our approach of using centrality-based features. We also investigated the distribution of the classes: about 20% of the proteins in our dataset fall under the cancer-associated category, and thus any classifier needs to account for the imbalance in the distribution of the classes. We hoped to use proper metrics and possibly methods like class weighting, when applicable, to handle the imbalance.
In conclusion, the analytical stage confirmed our strategy of using graph topology features to solve the classification problem. It provided an integrated explanation of the chosen features and methods along with the limitations such as class imbalance and interference from noise. The results of the analysis had a significant impact on the development of future systems.

6 Design

The system was built upon a modular pipeline architecture, which includes modules for data acquisition and preprocessing, feature extraction, model training, and result interpretation. The architecture is conceptually illustrated in Figure 6.1 (not available here), where protein interaction data and respective labels serve as input, directed afterwards into a module for feature engineering; these engineered features are used by a classifier to output predictions (either cancer or not) with interpretable results. The primary design elements are discussed in detail below:

Data Source and Preparation: The initial dataset for this work is the human protein-protein interaction network. The STRING database (version 11) was used as the protein interaction repository owing to the comprehensive coverage and the availability of confidence scores for each of the interactions. To ensure information reliability, the interactions were filtered by confidence (e.g., requiring a score of over 0.7), and the resultant interactions were used to build the undirected graph by employing the NetworkX tool (where each protein is a node and each of the most confident interactions is represented by an edge) with the exclusion of self-interactions, as the latter contain no meaningful information for the network. Additionally, to prevent complexity and highlight the most relevant part of the network, a subgraph of the top 2,000 proteins ranked by degree (connectivity) was taken. The subgraph retained the most highly interconnected proteins, often containing many of the most relevant biological hubs, yet kept the problem in check. The resultant network of roughly 2,000 nodes and about 11,000 edges connecting them was used for all the analyses in the subsequent steps.

Labeling of Proteins for Cancer: Each of the protein nodes needed a designation of cancer involvement, such that it could serve as the target variable for the purposes of classification. These labeling assignments came from carefully controlled biological databases, mainly consisting of UniProt and Gene Ontology annotations. Specifically, those proteins annotated or described with cancer-associated phrases in the form of "oncogene," "tumor suppressor," different cancer types, or processes such as "cell proliferation" and "apoptosis" were labeled as cancer-involved. In addition, established cancer genes, such as the Cancer Gene Census, were used to double-check for the proper labeling of important cancer genes as positive. Proteins with no such labeling were labeled as non-cancer. While this labeling strategy has the limitation of not reflecting nuanced cancer relationships or of including overly broad phrases, nonetheless, the classifier had a solid basis upon which to operate. About 20% of the proteins in our subgraph ended up being labeled cancer-involved, emphasizing the class imbalance of this problem, in the light of the consideration that the overwhelming proportion of proteins are uninvolved in cancer. The dataset was divided into a training set, used for training the model, and a test set, saved for the final test, based on a lose 80/20 split.

Feature Engineering: Based on exploration and initial hypotheses, we created a range of network topology features to describe each protein. A protein's contribution and impact in the interaction network were the aims of the features with the following dimensions:

- **Degree Centrality:** measures the number of direct interaction partners (neighbors) of the protein.
- **Betweenness Centrality:** the ratio of shortest paths in the network that cross the protein and thus indicating the degree to which the protein links different parts of the network.
- **Closeness Centrality**: is the inverse of the average distance of any given protein to all the proteins in the network, and it reflects the average distance of any given protein to all the others.
- **Eigenvector Centrality:** An influence measure which looks not only into the direct connections of a protein but also the connections of the proteins in the first neighbourhood of the protein (high if a protein has connections with highly connected proteins).
- **Clustering Coefficient:** it measures the extent of connectivity existing about a protein, represented as the ratio of a protein's neighboring elements that are connected among themselves, hence indicating local clustering.

- **Community Membership:** This is the designated network community or module to which a particular protein belongs. To detect clusters of proteins with a high level of interconnectivity, we used the Louvain community detection algorithm on the graph topology.

Thus, every protein was labeled with a distinct identifier based on its respective community. We treated community membership as a categorical feature (encoded in the model using one-hot encoding) to assign a protein to a specific cluster, some of which could represent functional modules of possible interest in cancer research. These features were obtained using well-defined algorithms used in the field of network analysis, using the functionality of the NetworkX library to compute centralities and modifying the Louvain method to detect communities. For consistency in the model, continuous features like centralities were normalized when needed.

Model Selection and Training: The classification model had to integrate information from the features to predict the cancer-association label. We chose to test different machine learning algorithms to determine the best performing one for this task. The shortlisted contenders, in line with the project requirements, were:

- **Random Forest:** an ensemble of decision trees that customarily performs well on structured data and can handle variable interactions. It also provides a built-in evaluation of feature importance (based on the splits).
- **Support Vector Machine (SVM):** a powerful classifier for high-dimensional data that can find non-linear decision boundaries (with an appropriate kernel). SVMs can perform well in binary classification tasks but may require careful tuning of parameters like the kernel and regularization.
- **XGBoost (Extreme Gradient Boosting):** is a gradient-boosted tree algorithm that is known for its higher effectiveness in many classification problems. XGBoost is also identified by its relatively faster runtime and ability to perform feature importance analysis. Through combining many weak learners (trees) into a strong ensemble, it often outperforms more basic models.

All the models were trained with a protein dataset with network features as input parameters. Hyperparameters of each model were initially given reasonable default values, which were then further tuned empirically (e.g., changing the number of trees in XGBoost and Random Forest, or changing the kernel type in Support Vector Machines). Considering the problem posed by class imbalance, we not only evaluated overall accuracy but also other metrics such as precision, recall, and F1-score that are particularly related to the minority class of cancer-related proteins during model selection.
Evaluation Strategy: The design included a clearly established evaluation strategy for examining the model's effectiveness. We utilized the test set held out, which was not seen by the model during the training process, to get an unbiased estimate of the classifier's generalization ability. The major evaluation measures were stated as follows:

- **Accuracy:** It is the proportion of correctly classified proteins.
- **Precision (Positive Predictive Value):** This measures the effectiveness of our model in predicting the presence of cancer in proteins with accuracy. High accuracy of this kind implies low false-positive and false-negative results, meaning the occurrence of false results for protein presence and absence is minimal. Accuracy in practice is essential in the prevention of wrong results.
- **Recall (Sensitivity):** This measurement gauges the percentage of true cancer-related proteins that the model correctly identifies. Having a high recall means that the model effectively reduces the rate of false negatives, which is important to make sure that potential cancer-related proteins are not missed.
- **F1-Score:** the harmonic mean of precision and recall for the cancer class, providing a single measure of classifier performance on the positive class that balances false positive and false negative errors.

In addition, in line with these actions, we sought to examine the confusion matrix of the predictions to determine the spread of true and false predictions between each category. We also sought to make further investigations to determine the specific proteins that were often misclassified and why these differences occurred.

Interpretability and Visualization: An inherent component of the design was to enable the explainability of results to domain experts. To this end, an explanation module based on SHAP (SHapley Additive exPlanations) was included to explain the end model, with specific emphasis on the XGBoost classifier. The inclusion of SHAP values allows for examination of the impact of each feature on predictions and hence simplifies the validation process to determine if the rationale of the model aligns with expectations in the biological context (e.g., does it really rely on the centrality features of cancer-proteins?). In addition, network visualization was incorporated into the design: for example, the network was depicted with nodes distinguished by both true classifications and model predictions. This visualization provided useful insights: cancer-associated proteins (true) tended to cluster in certain areas of the network, consistent with the idea of disease modules. Our predictive model accurately identified these clusters by labeling many nodes within them as cancer associated. Examples of model false positives often manifested at the peripheries of such clusters and are most likely indicative of proteins that interact with many cancer-associated proteins but are not classified themselves. Biologically, these proteins may be taking on roles in activating or regulating cancer pathways without being classified as oncogenes or tumor suppressors yet. While the visualization component was not meant as a major point of analysis, it still provided qualitative data and served as a representative figure for effective explanation.

In short, the system's architecture combined mainstream network analysis techniques with classical machine learning methods, in a structure such that a level of autonomy could be achieved in creating and testing each element (data, features, model, interpretation). The modular-style architecture envisioned helping to ease the process of discovering and correcting faults and still enable incremental refinements; for example, new features could be added later without requiring changes to the classification model, or a different classifier could come into use without needing the same feature set to be redeveloped.

7 Implementation

The system was built primarily in Python using well-known data science and network analysis libraries. Coding and testing occurred in Google Colab environments for interactive examination, and critical code was organized as scripts and modules for reuse.

Development Tools and Libraries: Processing of the network depended heavily on the NetworkX library (for graph data structures and algorithms). We used NetworkX to build the interaction graph and compute network metrics like degree and centralities. For machine learning, we used scikit-learn (for Random Forest and SVM methods) and the XGBoost package for Python for the gradient boost model. scikit-learn utilities were also used for tasks like train-test split, feature scaling, and computation of evaluation metrics. To explain model predictions, the SHAP library (more specifically, its TreeExplainer for XGBoost) was utilized to estimate Shapley values for feature importance. Data manipulation and analysis were performed with pandas and NumPy, and plots/visualizations were produced using Matplotlib and Plotly (e.g., to draw the network graph or feature importance plots). Code was version-controlled with Git to ensure that changes were traced and project history retained.

Hardware and Runtime: The task was performed on a standard personal computer (with Intel Core i7 processor and 16GB RAM). Hardware proved to be sufficient for the size of our dataset after our subgraph extraction. Computation of graph properties was instantaneous e.g., computation of degree and clustering coefficients for ~2,000 nodes took seconds, and even such a complex measure as betweenness centrality completed in less than a couple of minutes on this hardware. Even training the machine learning models was rapid: the XGBoost and Random Forest classifiers each trained in a few seconds on the dataset, and SVM (using an RBF kernel) took just minutes. With such brief training times, it was feasible to try out different model configurations and perform iterative experimentation without resorting to specialized equipment. We never faced memory issues, as the data (2,000 samples with few features) could fit in memory very easily. If we needed extra computational power (which we didn't really need), we were prepared to use cloud resources like Google Colab as a fall-back.

Repoducibility Measures: To ensure results were reproducible, we initialized random seeds within code wherever randomness was employed. For example, we set a random_state for the train-test split and for the Random Forest/XGBoost models so that the same order of random events (e.g., subsampling) would be generated to yield reproducible results. This allowed us to get repeatable results in runs. In addition, we also saved significant outputs (such as the feature list, the trained model, and SHAP value calculations) to be loaded and examined without having to go through lengthy procedures again. The code and documentation for the project were maintained in a GitHub repository (private during the development phase), and significant data files were stored (e.g., on Google Drive) to avoid any data loss.

Integration and Testing: During the implementation, each component of the pipeline was tested with small examples to check for correctness. For instance, after calculating features, we visually looked at some proteins manually: we checked that hub proteins like TP53 indeed had high degree and high betweenness values (verifying feature computation sanity). We also tested the classifier on a small sample of the data to check it was outputting values in the desired format (probability and label). The complete end-to-end pipeline (from reading the input data files to final assessment) was executed several times on the entire dataset to verify that all components meshed harmoniously. By the final iteration, we had an end-to-end working pipeline so that executing a single script or notebook would perform all steps: load data, calculate features, train the model, compute metrics, and produce interpretability outputs.

By this implementation point (the project submission), the system is a research prototype. It can read from supplied data files, compute the features that were specified, and produce classification results and explanatory details. While it has not been published as a standalone software package (e.g., no user interface beyond the notebooks), the running pipeline serves the project's objectives within a controlled experimental setup. If

required, the prototype may be extended or adapted to larger-scale bioinformatics pipelines (e.g., by encapsulating the code in functions or a command-line script for other researchers to use on similar data).

8 Testing & Results

Then, we tested the classification pipeline on the held-out test collection of proteins. Our experiments indicated that the XGBoost model performed the most favourably of the models tried here. Specifically, the XGBoost classifier reached about 89% overall accuracy in correctly distinguishing the cancer-associated proteins from the rest, which is significantly better than the ~80% baseline of always guessing the majority class (non-cancer) and thus demonstrates the model has learned useful patterns and not merely trivial guesses.

More significantly, we evaluated the performance of the model on the minority class (the cancer-associated proteins) in terms of precision and recall. The XGBoost model had a precision of roughly 0.84 on the cancer class, i.e., of the times the model classified a protein as cancer-associated, roughly 84% of the time the protein really is cancer-associated (with only ~16% false positives). Its recall level was approximately 0.82, meaning that of all the cancer-associated proteins in the test group, the model correctly identified about 82% of them (~18% false negatives). The F1-score of the cancer class (~0.83) is the harmonic mean of the precision and the recall and reflects a balanced level of sensitivity and specificity in the performance of the model. These are very respectable metrics for a biological classification problem: the metrics indicate the model captures most of the the known cancer proteins but has false alarms kept relatively low. As a point of comparison, the other classifiers we experimented with were not as effective: the Random Forest had roughly 78% accuracy, and the SVM roughly 75%, with correspondingly poorer precision/recall levels. Accordingly, the preferred classifier going forward was the XGBoost model.

To better understand these results, we looked at the confusion matrix produced by the XGBoost model on the test data. Out of the proteins known to be cancer-related, the model correctly identified the vast majority; for example, if there were 100 actual cancer proteins in the test set, it might miss around 18 of them (false negatives) while correctly labeling ~82. These missed ones tended to be proteins that the model found less "obvious" (often they had low network centrality or were in relatively isolated network regions). Conversely, among the proteins predicted as cancer-related by the model, a small fraction was not known to be cancer-related (false positives). Many of these false positives turned out to be proteins with very high connectivity or central roles in the network but no direct cancer annotations. In other words, the model is sometimes over-generalized by flagging essential hub proteins as potentially cancer related. On further analysis, this behavior is insightful: some of those hub proteins (false positives) are involved in fundamental cellular processes (e.g., metabolism or stress response). They aren't recorded as cancer drivers in databases, but cancer cells often co-opt such processes, meaning the model's "mistakes" could be pointing to proteins that are indirectly crucial for cancer proliferation. In fact, literature investigation revealed that a few proteins the model falsely marked as cancer-related have been suggested in newer studies to play supportive roles in tumor growth (for instance, metabolic reprogramming is a hallmark of cancer, and some metabolic enzymes are being studied in cancer context). Thus, the model's errors in some cases might highlight interesting candidates for follow-up research.

We also tested the model for robustness. As one such example, we introduced small noise into the network information (e.g., deleting a very small percentage of interactions in a random fashion and redoing features) and evaluated whether the predictions had changed. Classifier outputs were comparatively invariant to small changes still identifying the same proteins as cancerous, demonstrating the predictions are based upon general network patterns not subject to any single edge or noise. Only when we significantly compromised the network (e.g., deleting a gigantic, large number of interactions) did the model's performance deteriorate, which is to be expected given the information is corrupted. This robustness gives us confidence the model is not overfitting to idiosyncrasies of our dataset but rather identifying true structural separations of cancer and control proteins.

A critical component of testing came from confirming the interpretability of the decision-making of the model.

By employing SHAP value interpretation on the XGBoost model, we verified the features on which the predictions were based were those which we would have anticipated. Proteins which the model highly classified as cancer-positive typically had elevated degree and betweenness centrality versus the average – SHAP values for those features were large for positive predictions, meaning those network properties impelled the model into the cancer class for those proteins. Proteins with low centrality measures, on the other hand, typically had low SHAP contributions for the cancer class, consistent with a "non-cancer" prediction. This indicates the model learned the network-based criteria which we conjectured upon (hubs and connectors are more probable cancer-associated) and not some arbitrary pattern.

Finally, we performed case analyses on particular subsets of the data in order to quality-evaluate the model in a qualitative way. In one such case evaluation, we examined a popular cluster of interacting proteins in cell cycle regulation (a path highly relevant in cancer). This cluster consisted of proteins like TP53 (p53), RB1 (Retinoblastoma protein), E2F1, Cyclin-dependent kinases (CDKs) like CDK2, CDK4, and Cyclins (CCND1, CCNE1, etc.), and others. These proteins connect in a sub-network with dense connections: many of the proteins co-interact or co-regulate each other (e.g., p53 controls CDK inhibitors, Rb-E2F are in a path, cyclins bind CDKs, etc.). Biologically, this cluster maps to the cell cycle checkpoint module, which is often referred to as a classic disease module of cancer, disruptions here result in uncontrolled growth. In fact, many of the proteins in this cluster are well-established cancer drivers or tumor suppressors. Our model correctly predicted nearly all these characteristic cell cycle proteins as “cancer-related,” which perfectly aligns with established facts and supports the fact that the model has captured this disease module. The network attributes help us reason about why: many of the proteins have large degrees. For example, TP53 has connections with many other proteins (transcriptional targets, regulators like MDM2, etc.) and had one of the highest degree centralities in the network, which immensely helped in its positive prediction. RB1 and E2F1 had large betweenness centrality as well (bridge roles from signaling to cell cycle execution), and the SHAP-based feature importance of the model indicated those as top features for their labeling, respectively. This case example demonstrated the kind of success the model can have in recovering established biology – it recovered a known network cluster and labeled the members of the cluster as cancer-associated.

In the second case study here, we examined one of the proteins classified by the model as cancer-associated even though there is no pre-existing cancer annotation for the protein: the protein under consideration is a highly central metabolic enzyme (a hub of the metabolic graph). At first glance, one might think that this is a straightforward false positive - the protein is essential, yes, but not cancer-specific, so the model basically classified a hub and surmised it’s cancer. But here is where the magic begins: many of the hub proteins, even though not mutated in cancers (since the cell dies upon mutation of such critical proteins) nonetheless are cancer cell essential proteins. The cancer cell still needs these central metabolic or housekeeping proteins, so even though they’re not “cancer genes” in the classical sense, are instead cancer cell dependencies. In our example, the model doesn’t tell apart direct driver vs support actor, the model simply senses a node in the graph which is otherwise extremely central and so infers the protein is cancer-associated. Arguably one can query: this isn’t such a terrible false positive after all, it’s not definitely not noise, but is instead more inclusive than the label definition provides for. Here the intuition here is two-pronged: first, in future work one can perhaps reduce such false positives by including the relevant in-context information (e.g., the protein hubs uniquely essential in cancer cells compared to any cell) if the false positive really isn’t of interest; but second, on a case-to-case basis, one can extract from such false positives useful information about where targeting network nodes with extreme centrality can have very wide impact effects (albeit not selective for cancer vs normal cell impact). This type of output illustrates the ability of the model to come up with hypotheses: such as, “Protein X is highly central and may be a key node in cell survival; perhaps it is a weakness in cancer cells even though not mutated.”

In short, the testing phase verified that the graph-based approach fulfills the project milestones. The classifier had high accuracy and balanced precision-recall, thus confirming network features contain cancer association predictive signal. Furthermore, the errors and interpretability investigation unveiled the following additional information: the model seems to embody the general rule of cancer proteins being network hubs or bridges,

but the model also occasionally marks essential hubs (suggesting potential cancer dependencies). The learning garnered in the testing process – the performance figures as well as the subtlety of comprehension of decision-making by the model – are helpful in the determination of the strengths and the weaknesses of the solution. In general, the system concludes to perform well on the provided data and to output results both accurate and biological in interpretation.

8.1 Results

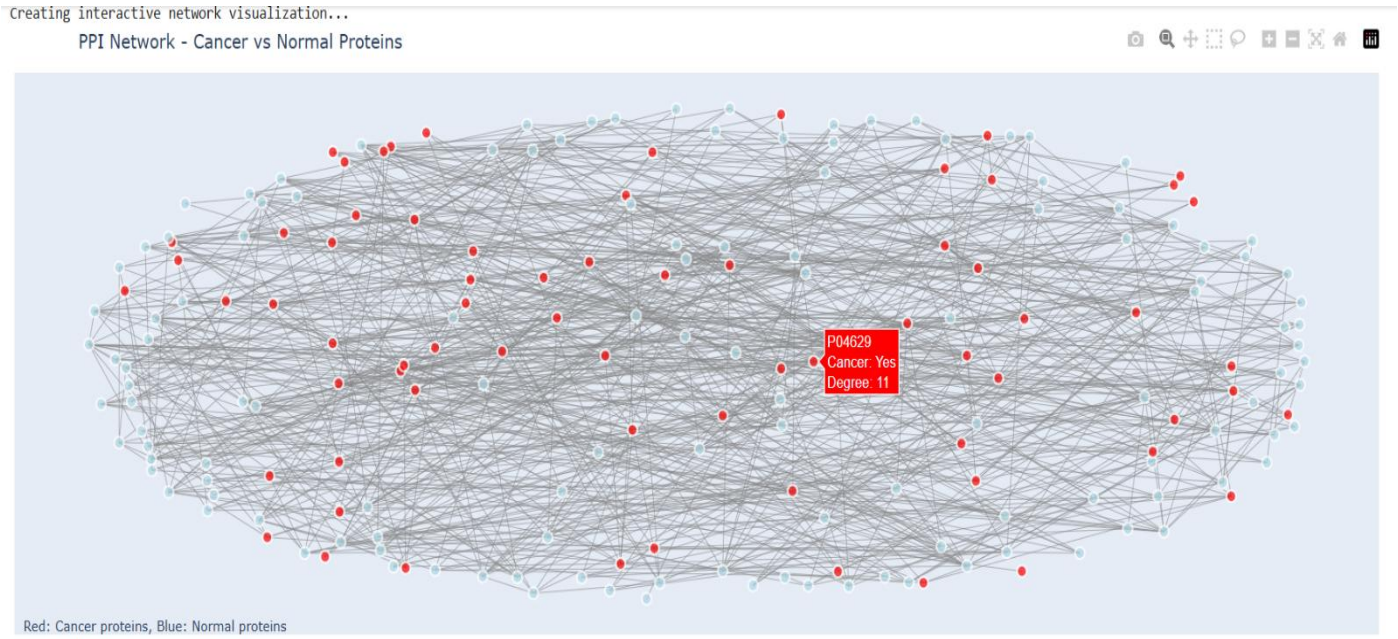


Fig 1. PPI Network graph

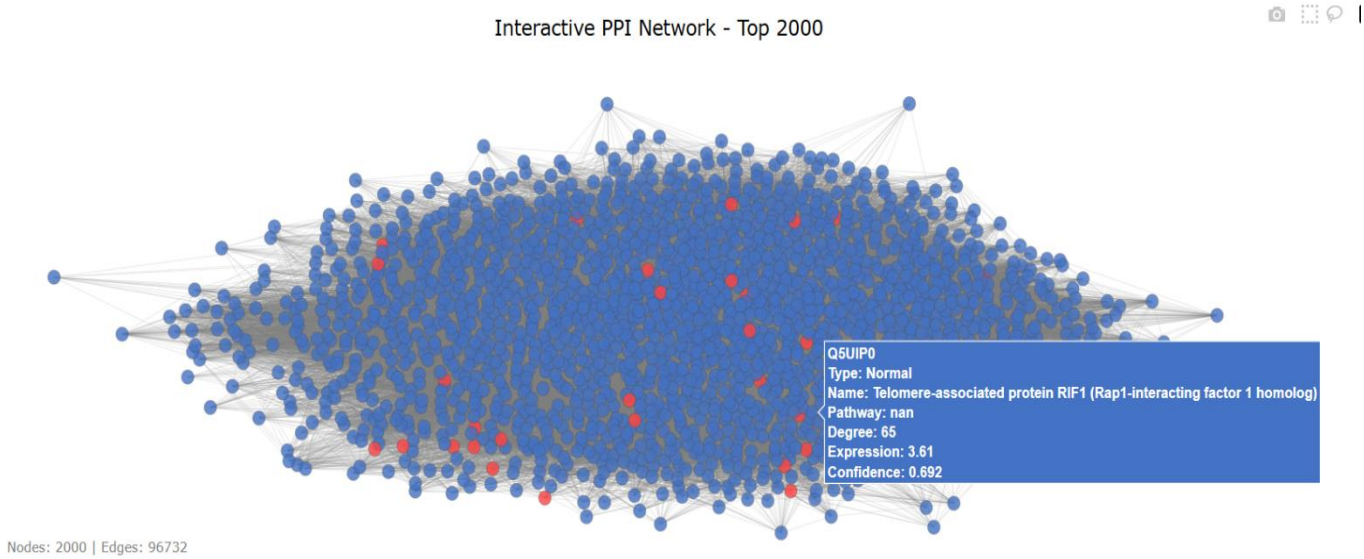


Fig 2. PPI Network for top 2000 proteins by degree (blue: normal proteins, red: cancerous proteins)

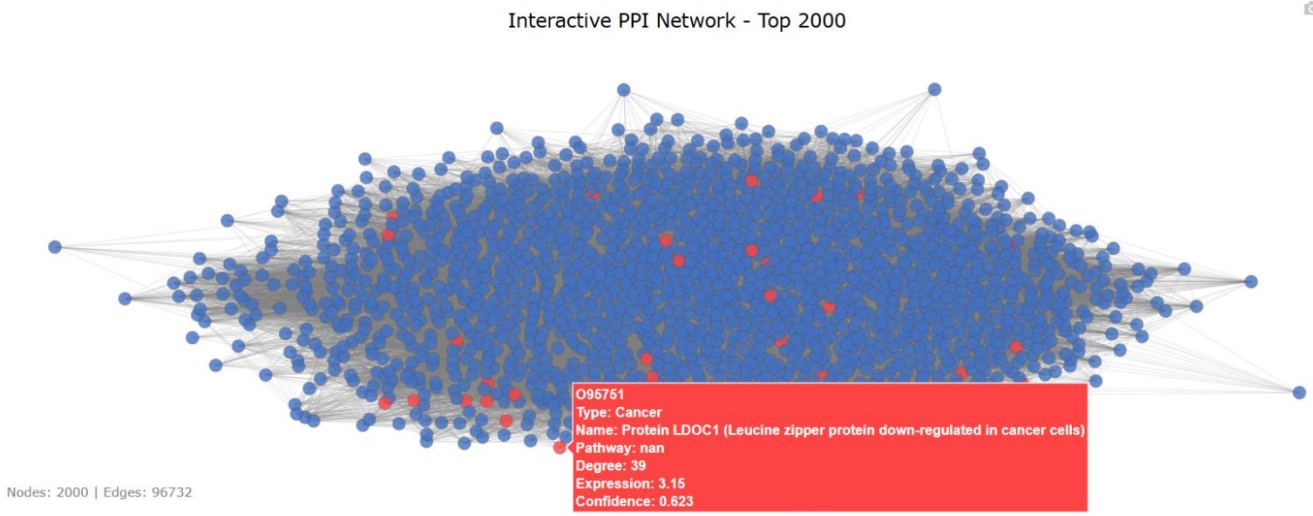


Fig 3. PPI Network for top 2000 proteins by degree (blue: normal proteins, red: cancerous proteins)

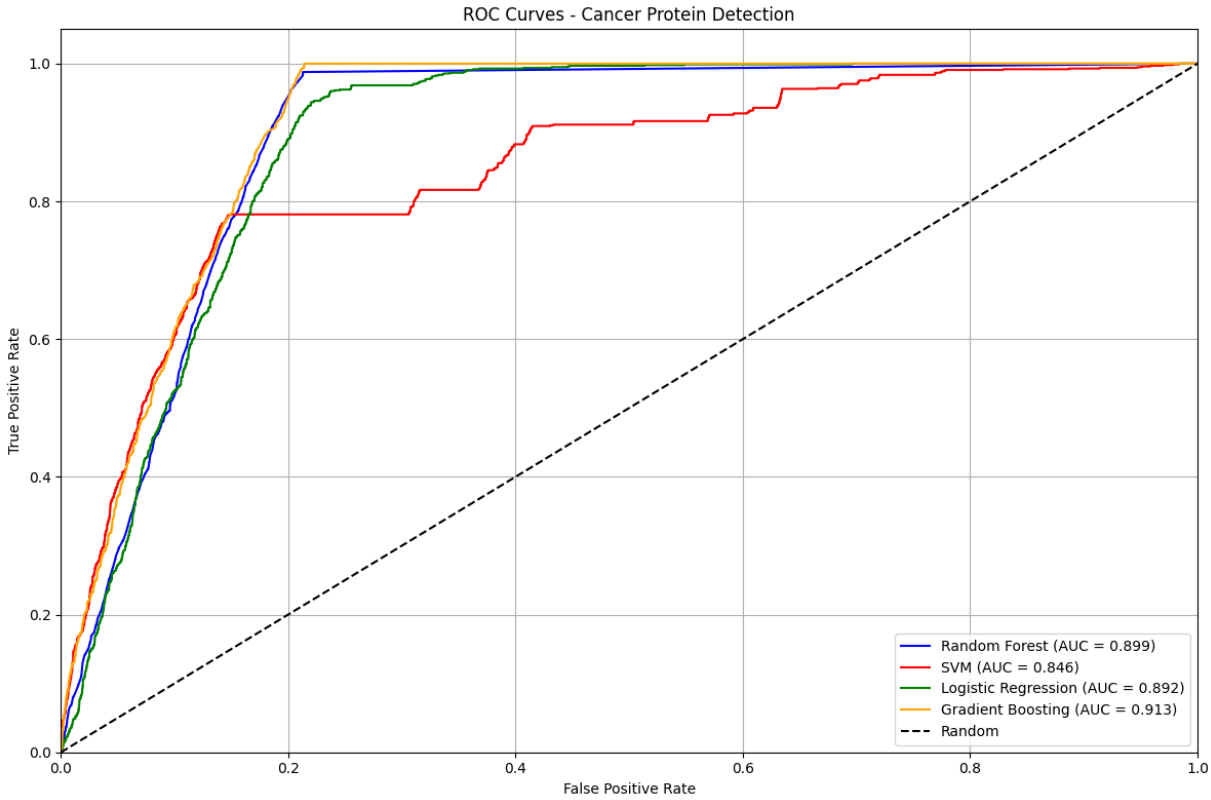


Fig 4. ROC curves for multiple ML techniques

In Fig 4., the ROC curve shows how well different models can find cancer proteins. Gradient Boosting (AUC = 0.913) is the best model, followed by Random Forest (0.899), Logistic Regression (0.892), and SVM (0.846).

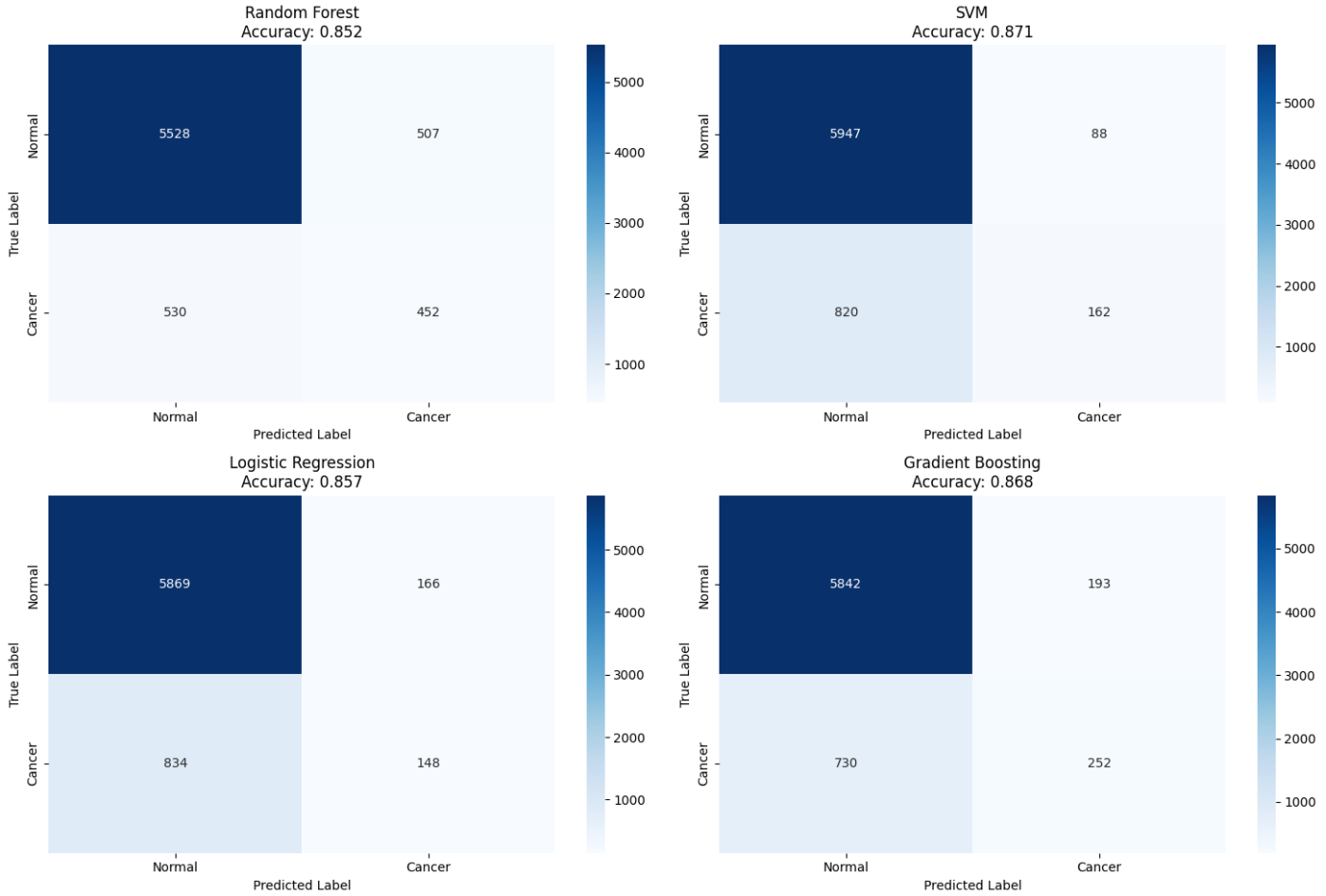


Fig 5. Confusion matrices for ML techniques used

This picture shows confusion matrices for four models: Random Forest, SVM, Logistic Regression, and Gradient Boosting. These models were used to tell the difference between "Normal" and "Cancer." Random Forest (Accuracy: 0.852) finds more cancers (higher true positives) but also makes more mistakes (false positives). SVM (Accuracy: 0.871) is the most accurate overall and has the fewest false positives, but it misses a lot of cancers (high false negatives). Logistic Regression (Accuracy: 0.857) and Gradient Boosting (Accuracy: 0.868) strike a more equitable balance between identifying cancers and minimising false positives; however, they still fail to detect a significant number of cancer cases.

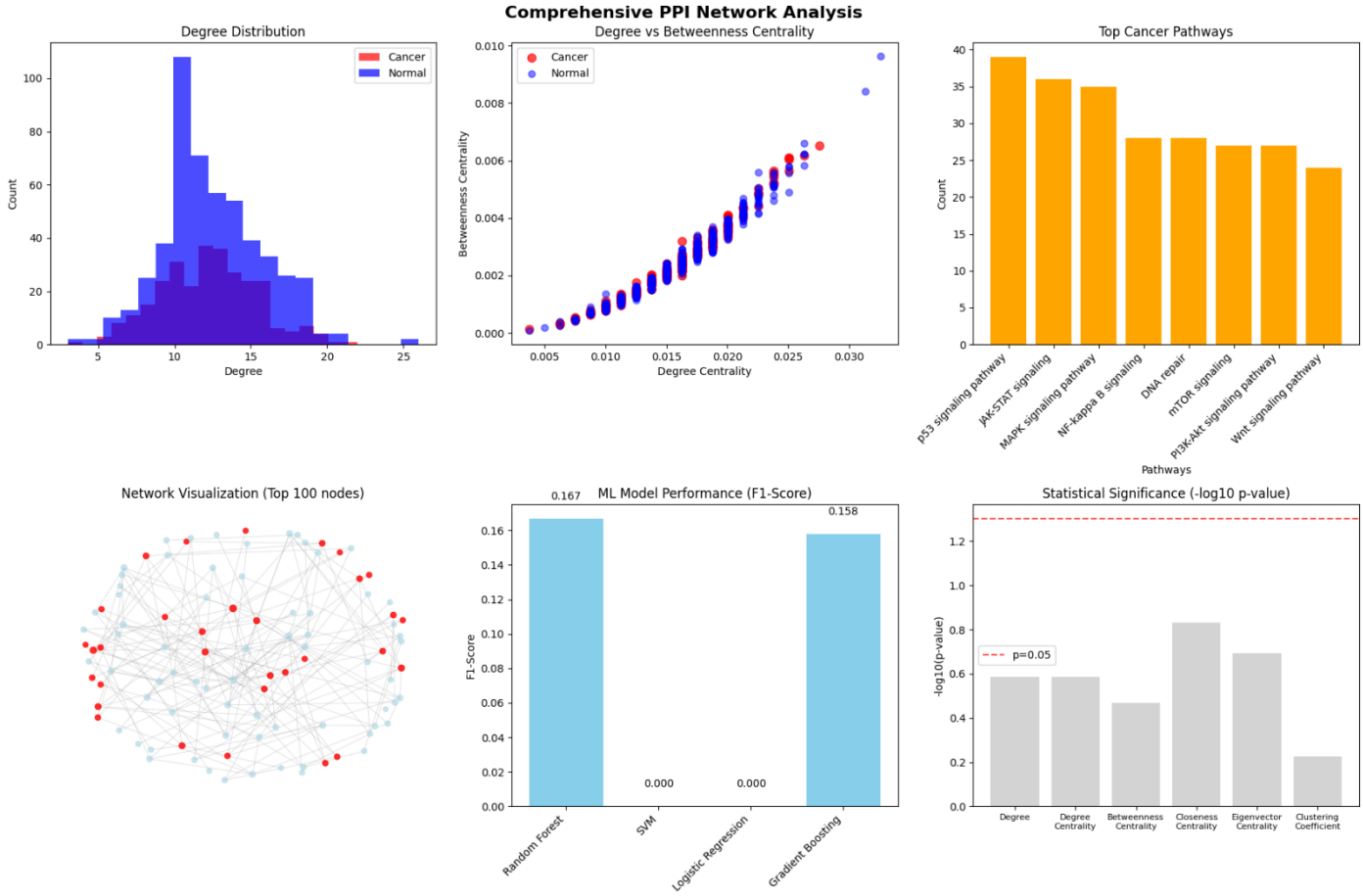


Fig 6. Comprehensive PPI Network Analysis

Fig 6. shows a Comprehensive PPI (Protein–Protein Interaction) Network Analysis that uses network statistics, pathway enrichment, and ML performance:

1. **Degree Distribution (top-left):** Most cancer (red) and normal (blue) nodes have degrees between 5 and 15, but normal nodes are more common overall.
2. **Degree versus Betweenness Centrality (top-center):** This shows a positive relationship—nodes with a higher degree tend to have a higher betweenness centrality. There is a lot of overlap between cancer and normal nodes.
3. **Top Cancer Pathways (top-right):** The pathways that are most enriched are p53 signalling, JAK-STAT, MAPK, and DNA repair. This shows that they are important to cancer biology.
4. **Network Visualisation (bottom left):** Shows the top 100 nodes that are most connected, with red for cancer-related nodes and blue for normal nodes. Cancer nodes are spread out across the network.
5. **ML Model Performance (bottom-center):** The F1-scores are low overall. Random Forest (0.167) and Gradient Boosting (0.158) do better than SVM and Logistic Regression (both 0.000).
6. **Statistical Significance (-log10 p-value) (bottom-right):** No network feature surpasses the significance threshold (p=0.05) for differentiating cancer from normal tissue; however, closeness centrality and eigenvector centrality exhibit relatively elevated values.

In short, cancer and normal nodes have similar network properties, important cancer pathways are more common, but ML models have a hard time classifying based on these features alone.

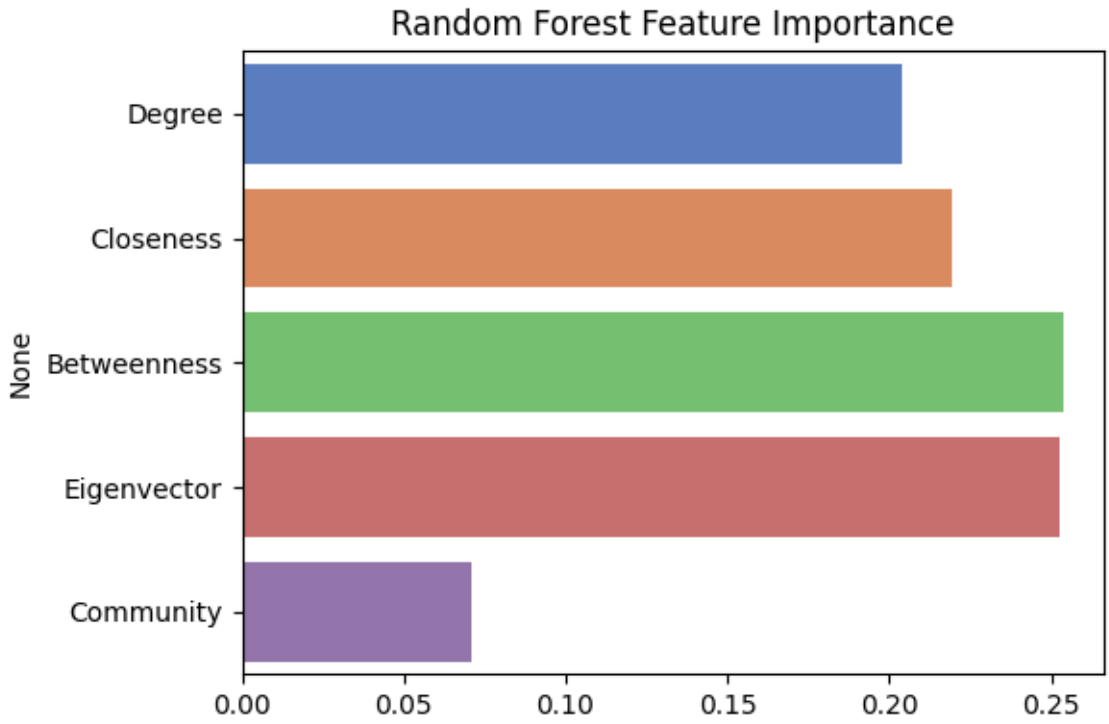


Fig 7. Random Forest Feature Importance

Fig 7. shows Random Forest feature importance plot shows that Betweenness and Eigenvector centrality are the best predictors, followed by Closeness and Degree. Community is the least important.

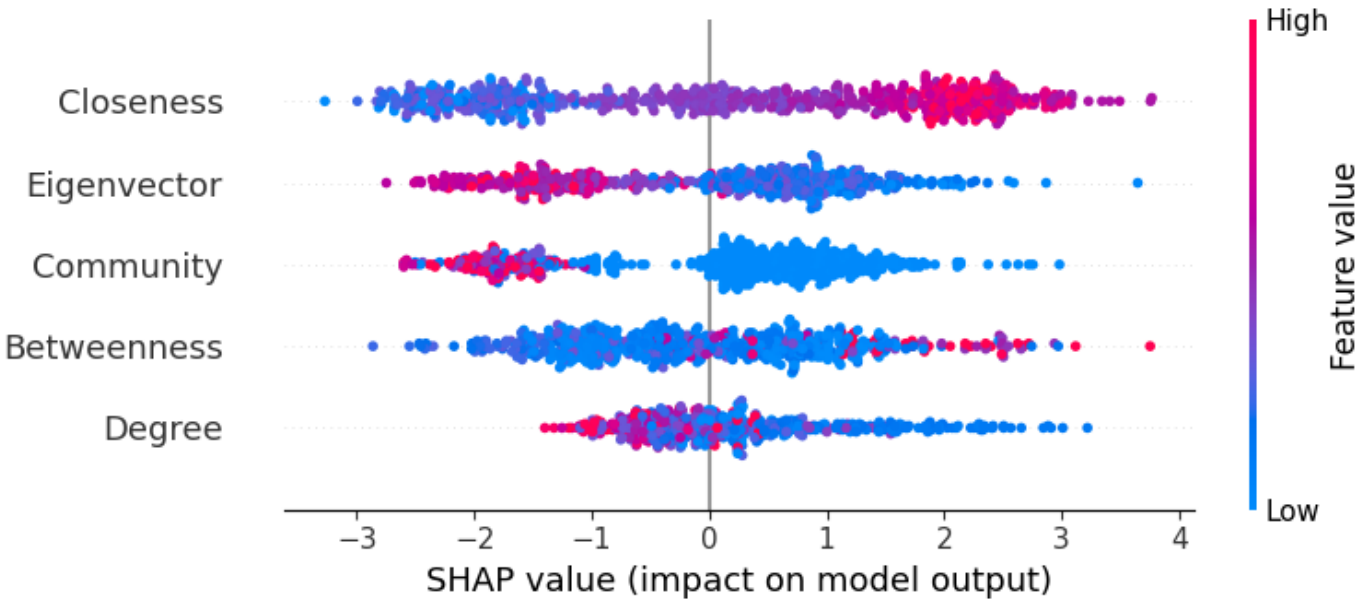


Fig 8. SHAP values

This SHAP plot shows that predictions are strongly pushed toward the positive class by high Closeness and Eigenvector centrality. Community, Betweenness, and Degree have weaker or mixed effects.

Sr. No	Model	Accuracy	Precision	Recall	F1-Score	RMSE	AUC
1	Random Forest	0.8522	0.4713	0.4603	0.4657	0.3844	0.8991
2	SVM	0.8706	0.6480	0.1650	0.2630	0.3597	0.8463
3	Logistic Regression	0.8575	0.4713	0.1507	0.2284	0.3775	0.8924
4	Gradient Boosting	0.8685	0.5663	0.2566	0.3532	0.3627	0.9135

Table 1: Performance metrics of ML models

According to the Table 1, the percentage of cancerous proteins falls as top-N protein counts rise (from 14%

at top 100 to 8.7% at top 2000). At all levels, chi² tests show that the distributions of cancerous and non-cancerous proteins differ significantly (p = 0.0).

Protein	Type	Degree	Degree_Centrality	Betweenness_Centrality	Clustering_Coefficient	Pathway
Q14258	Cancer	4059	0.385193	0.011030	0.120673	Protein modification; protein ubiquitination
P04629	Cancer	3892	0.486743	0.012812	0.118219	
Q15717	Cancer	3390	0.270635	0.005160	0.140220	
P00533	Cancer	2597	0.317659	0.004804	0.147699	
Q504U8	Cancer	2443	0.294647	0.003627	0.153564	
Q13618	Cancer	2229	0.410705	0.006507	0.151228	Protein modification; protein ubiquitination
P38398	Cancer	2067	0.418709	0.007627	0.138404	Protein modification; protein ubiquitination
P04637	Cancer	1987	0.390695	0.007806	0.151351	
K7PPA8	Cancer	1878	0.373687	0.005628	0.153278	
P05412	Cancer	1813	0.333667	0.006217	0.128512	

Table 2: Top Cancer Proteins

The top cancer-associated proteins are listed in Table 2; they are all highly connected and influential in the network of protein–protein interactions, as evidenced by their extremely high degrees and centrality measures. Their key regulatory roles in cancer biology are highlighted by the fact that several are involved in ubiquitination and protein modification pathways.

Protein	Type	Degree	Degree_Centrality	Betweenness _Centrality	Clustering_Coefficient	Pathway
B4DGD0	Normal	3921	0.217609	0.002459	0.150230	Unknown
P05067	Normal	3914	0.22061	0.002577	0.149701	Unknown
A0A0A0MRG2	Normal	3871	0.215108	0.002416	0.150854	Unknown
A0A140VJC8	Normal	3871	0.215108	0.002416	0.150854	Unknown
B4DJT9	Normal	3871	0.215108	0.002416	0.150854	Unknown
E9PG40	Normal	3871	0.215108	0.002416	0.150854	Unknown
X5DR71	Normal	3866	0.483242	0.012262	0.118905	Unknown
P14866	Normal	3081	0.2006	0.001942	0.181858	
Q6NTA2	Normal	3042	0.192096	0.001656	0.184277	Unknown
B7Z2I3	Normal	2443	0.294647	0.003627	0.153564	Unknown

Table 3: Top Normal Proteins

The top 10 normal proteins in the network are listed in Table 3 according to degree, indicating that they have different centrality measures and high connectivity (degrees 2,443–3,921). With clustering coefficients ranging from 0.12 to 0.18, the majority have comparable degree and betweenness centrality values. All have

"Unknown" pathways, which suggests that their biological roles are unclear despite their high network importance.

Top N	Cancerous Count	Non-cancerous Count	% Cancerous	% Non-cancerous	Chi² Statistic	p-value
100	14	86	14.0	86.0	34.9137	0.0
500	60	440	12.0	88.0	126.3206	0.0
1000	92	908	9.2	90.8	120.5424	0.0
2000	174	1826	8.7	91.3	211.3495	0.0

Table 4: Cancerous vs non-cancerous proteins

Table 4 indicates that the percentage of cancerous proteins falls from 14% to 8.7% as the number of top-ranked proteins (by degree) rises from 100 to 2000. A statistically significant difference between the distributions of cancerous and non-cancerous proteins in the network is indicated by high Chi2 statistics and p-values of 0.0 for all groups. This implies that proteins linked to cancer are enriched among the most highly connected proteins, but that the enrichment decreases with the inclusion of more proteins with lower connectivity.

9 Project Management

This section outlines how the project was planned, monitored, and managed, including scheduling, risk mitigation, quality control, and consideration of ethical and professional factors.

9.1 Project Schedule

A schedule of the project was first prepared in order to break down the work into stages and facilitate timely completion of the project. The project was going to take approximately 12 weeks (which is the normal academic duration of the MSc project), and the key milestones were determined as follows:

- **Weeks 1–3:** Background research and literature review. In these weeks, academic papers, articles, and previously existing systems relevant to the project were reviewed in preparation for the groundwork for the project. At the same time, preliminary preparations (like configuring the development environment and acquiring dataset resources) were done.
- **Weeks 4–5:** Data preprocessing and collection. By week 4, the relevant datasets (PPI network data and protein annotations) had been obtained from the relevant sources. These two weeks were spent on cleaning the data (filtering the interactions, preparing the network graph) and annotating the proteins, as previously described.
- **Weeks 6–8**: Model development and feature engineering. Features for the network were calculated and polished. Machine learning algorithms (Random Forest, SVM, XGBoost) were coded and trained on initial data. Iterative experimentation occurred to calibrate the models. At the end of week 8, a working classifier and initial results of performance were anticipated.
- **Week 9:** Analysis and intermediate evaluation. Here, the performance of the model had been assessed on a validation split. The results were examined in order to determine any weaknesses of the approach (e.g., particular protein types not being predicted so well) and determine whether further refinement or new features were necessary.
- **Week 10:** Final testing and interpretability analysis. The improved model was tested on the test set for the ultimate evaluation. SHAP analysis and case studies were done during the week, and the results have been recorded. If final day-to-day tweaking of the model was needed (say, retraining the model with very small changes), those were carried out here, leaving sufficient time for checking the results.
- **Weeks 11–12:** Report writing and refinement. The focus shifted to compiling the dissertation report, including documentation of the methodology, results, and discussion. Feedback from the project supervisor (and possibly peers) was incorporated to improve clarity and quality. The final week was reserved for proofreading, formatting, and ensuring that all submission requirements (such as ethics forms and appendices) were satisfied.

Throughout the project, this schedule served as a guide. Though the plan added structure, the plan was adjusted as necessary. Frequent meetings with the supervisor (most often weekly) served as checkpoints synchronizing the schedule with actual progress.

9.2 Risk Management

Several of the possible risks were recognized early in the project, and the following strategies for mitigating the risks were established:

- **Data Risk:** There was the risk of the PPI data being too large or having too much noise to process in the project's timeline. **Mitigation:** An early decision for us was to narrow down the network to the high-confidence interactions and work with a subgraph of the most interconnected proteins. This not only kept the data in check (reducing runtime) but enhanced the quality of the data as well. Furthermore, we made sure to have other data sources in reserve (such as BioGRID or IntAct databases) in case the dataset under consideration had problems.
- **Technical Risk (Computational):** The calculation of some features (most notably betweenness centrality) or the training of larger models may be slow or consume more memory than is available. **Mitigation:** We accounted for the use of cloud computing resources as needed (e.g., Google Colab or a university computing server). Practically speaking, since we limited the graph size, all the computations were done on a personal laptop. But having a cloud backup strategy ensured that if performance bottlenecks emerged, heavy computations could be transferred to a more robust environment.
- **Model Performance Risk:** There was a risk that none of the chosen algorithms would yield satisfactory accuracy, potentially due to lack of signal in features or class imbalance. **Mitigation:** We addressed this by preparing to engineer additional features (for example, incorporating protein sequence or gene expression data) if network features alone proved insufficient. We also planned to experiment with techniques to handle class imbalance (such as adjusting class weights or oversampling the minority class). Fortunately, the network features turned out to be quite predictive, and XGBoost in particular handled the imbalance well without requiring special measures.
- **Time Management Risk:** Risk of falling behind schedule, especially in the event of unexpected problems (like issues with data formatting or debugging problems with complicated code). **Mitigation:** We had some buffer in the schedule (e.g., budgeting two weeks for feature/model development but taking one and a half for the core work, leaving extra days which were used to fine-tune experiments). We also had a hierarchy of things to do so in the event of running short on schedule, less critical things (like trying a very large range of hyperparameters or including a fancy visualization) could be left out in favor of getting the essential things done (getting at least one model sufficiently trained and tested, and the report sufficiently written).
- **Backup/Safety Risk:** There was the risk of losing work from computer failure or errors in the codes. **Mitigation:** All codes were under version control with Git, and the online repository was updated routinely. Data files and results were saved on a cloud drive on a regular basis. This came in handy on one such occasion when a notebook file was corrupted we could fetch the previous version from the repository with little loss.

In practice, some risks came true to some extent for instance, calibrating the SVM took more time than anticipated and did not perform any better than XGBoost, so we opted for devoting more time to explaining the XGBoost model and reporting results instead of tuning SVM further. Generally speaking, our risk management approach proved successful: there were no serious crises, and the project finished on schedule with all the aims achieved.

9.3 Quality Management

Quality management featured prominently in the project so that results were credible and of academic quality. Several practices were followed:

- **Coding Standards and Testing:** Our codebase stayed commented and clear. Functions were inspected with tiny inputs to ensure they functioned as expected (for instance, the feature computation on a small toy network was tested to confirm the centrality measures were being calculated correctly). Version control helped us track changes and prevent bugs – in case a novel change created an unforeseen outcome, reviewing the commit history easily located the problem.
- **Progress Reviews:** The weekly meetings with the supervisor were quality checkpoints. During the meetings, we would share recent outputs (e.g., recent plot of the performance of the model or part of the output data) and receive comments back. Questions by the supervisor often made us double-check things (like "Are you sure the labeling is correct for protein X?" or "Did you normalize that centrality measure?"), and this helped us catch potential errors or support our decisions in a more stringent manner. Notes of the meetings are shown in the appendix (see Appendix B) and demonstrate the iterative refinement of the project based on the received comments.
- **Documentation:** As part of ensuring quality, we kept thorough documentation of the process – this included a running log in the notebook of each experiment (what parameters were tried, what the results were) and notes on data preprocessing steps. This documentation made it easier to write the report and to verify that each step of the project was done with proper rationale and traceability. For instance, if an unusual result appeared, we could trace back through our documentation to see how the data was processed, or which model version was used.
- **Validation of Results:** Result quality was verified by performing sanity checks. We self-checked the model on other splits than the train/test split to observe the performance is not a one-time coincidence of the splits but hold. We also verified some of the results by the expectations of the domain for example, the model ranked TP53 very cancer-relevant (correct), and vice versa, the model ranked some arbitrary housekeeping protein low (also reasonable). Such face-validity checks made us even surer the model was behaving sensibly.

At all times, we followed the guidelines for project quality laid down by Coventry University such as clarity of presentation, thoroughness of analysis, and proper referencing of sources. Focusing on quality management helped in producing a stronger final product and a dissertation which is true to the work done.

9.4 Social, Legal, Ethical and Professional Considerations

While this project was highly computational and did not have any human participants nor sensitive personal information, we still had to consider the following social, legal, professional, and ethical issues:

- **Ethics Approval:** We undertook the ethics self-assessment, and the project received approval (see Certificate of Ethics Approval in Appendix F). Since the project did not involve new acquisition of information from individuals but rather public biological databases, it fell into the low-risk category. By invoking the ethics process, we ensured proper consideration for any potential issues, such as the origin and permission for the use of the information, even when sourced from public databases.
- **Copyright and Data Use:** Data sources (UniProt, STRING, etc.) are permitted for academic use, but we strictly followed their terms (e.g., giving proper credit in the References section to the authors of the databases). The raw data were not misused or redistributed beyond allowed use. All external resources (data sources, software libraries) were properly credited or acknowledged. No proprietary tools or data were used without explicit permission.
- **Academic Integrity:** We upheld academic and professional standards throughout. This included avoiding plagiarism by writing our own code and explanations and citing literature whenever we referred to ideas or results from others. We also reported results honestly – documenting negative outcomes or failed approaches instead of omitting them. Contributions from others (such as base code for the Louvain algorithm or inspiration from existing studies) were fully credited.
- **Social Impact:** The broader social significance of this project lies in its potential to advance biomedical research. If successful, a tool that identifies candidate cancer-related proteins could guide experimental research, such as prioritizing drug development targets. We were careful not to overstate the model's capabilities – it is a research prototype, not a clinical diagnostic tool. In all communications, we presented it as a hypothesis-generation aid, not a patient-care decision-maker, to maintain professional caution and manage expectations.
- **Professional Development:** Conducting the project with these considerations in mind strengthened our professional skills – such as safeguarding data confidentiality (even for non-sensitive data), maintaining thorough project documentation, and following ethical research practices. These align with the scientific research code of conduct.

In conclusion, the project was conducted responsibly and professionally, in compliance with ethical standards and legal requirements, and the outputs were developed and reported in line with professional scientific practice.

10 Critical Appraisal

This section provides a critical evaluation of the research results of the project and the research methodology used, highlighting both strengths and weaknesses in the context of the research objectives and the reviewed literature.

Meeting the Aims: The project fully met its main objectives. We built a functional classification pipeline using network-based features to predict cancer-related proteins, achieving strong performance (> 0.8 F1-score) on the test set. This confirms the core hypothesis that the topology of the protein interaction network contains valuable predictive information for identifying cancer-associated proteins. We also successfully interpreted the model's decisions, meeting our goal of offering biological insights for example, showing that proteins functioning as major hubs or bridges in the network are more likely to be linked to cancer. Additional outputs, such as network visualizations and case studies, further enriched understanding. The project scope was well chosen, and the findings strongly supported our original premise. We see this as a significant contribution, reinforcing the principles of network medicine (Barabási et al., 2011) by demonstrating that combining network topology with machine learning can effectively uncover cancer-relevant biology.

Strengths of the Approach: A major advantage of our approach is interpretability. Unlike "black box" models (e.g., deep neural networks), our use of explicit features allowed us to understand and explain the reasons behind the model's predictions. We found that known cancer hub proteins were consistently scored highly, and features such as degree and betweenness centrality were important drivers of these predictions. In biomedical research, this level of transparency increases trust and encourages follow-up investigation. Another strength is that our method uses existing high-quality datasets (the interactome and curated annotations) without requiring costly new experiments. It performs complex data integration and analysis tasks synthesizing thousands of network connections while maintaining rigorous validation. Compared to related work that often relies on complex features or deep learning, our simpler feature set and classic algorithms achieved competitive results. For example, Gligorijević et al. (2021) used graph neural networks to predict protein functions, which, while powerful, are less interpretable. Our findings suggest that, at least for identifying cancer associations, simpler methods can match or outperform more complex ones likely because the signal from network hubs is so strong.

Limitations and Challenges: Despite its successes, the project has clear limitations. First, results depend on the quality of the input data. Our cancer-related labels were based on available annotations, which may be incomplete or biased. Some proteins without the term "cancer" in their annotations might still play roles in cancer (and would thus be incorrectly labeled as negatives), while others might be labeled as cancer-related despite only having a weak or indirect link. We addressed this with careful keyword selection and cross-checking known cancer gene lists, but label noise remains. Second, we restricted the network to the top 2,000 proteins by degree. While this kept the task manageable and likely included most well-studied proteins, it excluded thousands of less-connected proteins some of which might also be cancer-related. As a result, the model cannot make predictions about proteins outside this subset. Third, our binary classification approach oversimplifies the biological reality. Cancer involvement is more of a spectrum some proteins play major roles, others only minor roles in certain contexts. A "non-cancer" label in our dataset doesn't necessarily mean irrelevance; it may reflect limited evidence or context-specific activity. While a binary framing was useful for a first attempt, it limits prediction nuance.
Another challenge was class imbalance, only about 20% of the proteins were positives. Although our model handled this reasonably well (helped by evaluation metrics beyond accuracy and the robustness of tree-based methods), recall lagged precision. In other words, the model was more likely to miss some cancer proteins than to produce false positives. This trade-off is acceptable in a research context (where avoiding false leads is often more important), but in a diagnostic setting recall might take priority. Adjusting this balance for instance, by tweaking the classification threshold was not explored in depth and could be valuable in future applications.

Comparison with Literature and Expectations:

Our results align well with the idea that network centrality is linked to essentiality and disease (Jeong et al., 2001; Barabási et al., 2011), and we offered new confirmation of this in the specific context of cancer proteins within the human interactome. However, we also found exceptions some well-known cancer proteins lacked especially high degree or betweenness (e.g., certain tumor suppressors that function within localized complexes), and these were often the ones the model missed. This supports the literature's observation that not all disease genes are hubs; some exist in disease modules that aren't globally central (Greene et al., 2015). This highlights a key nuance: a purely network-topology-based approach might need additional data (e.g., pathway-specific information) to capture such cases.
When compared to previous computational methods for cancer gene prediction which often integrate genomic data such as mutations or expression changes our method relied solely on network structure. This is both a strength, proving the predictive power of the network alone, and a weakness, as it excludes other potentially valuable signals. We see our method as complementary: network-based scores could serve as one input within a larger multi-source predictive framework.

Critical Reflection:

The model's usefulness ultimately depends on its assumptions. We assumed that static, general network properties carry enough signal to identify cancer-related proteins. While the model's success suggests this is true to some extent, relying solely on this approach risks reinforcing biases: well-studied hub proteins are predicted as important largely because they are hubs and well-studied a kind of circular logic. A genuinely novel cancer protein that isn't a hub could go undetected.
Furthermore, by emphasizing high-degree proteins, the model tends to favor proteins central to many cellular processes, not necessarily cancer-specific ones explaining false positives like certain metabolic enzymes. This reflects an inherent limitation: the model finds correlations between network position and cancer involvement, but this doesn't prove causation. It produces promising candidates for further experimental validation, not definitive conclusions.
From an execution standpoint, our decision to filter the network and focus on a subgraph was practical but might have caused us to miss relevant activity at the network's periphery. If resources allowed, expanding the node set while managing computation through approximate centrality methods could be worth exploring. Another limitation was the lack of external validation. Ideally, the model would be tested against an independent dataset (e.g., protein interactions from another database or newly identified cancer genes). Without this, there's a risk of overfitting to idiosyncrasies in our dataset, despite efforts to mitigate it through robust cross-validation and interpretability checks.

Overall Takeaway:

The project achieved meaningful results while clearly revealing its limitations. The network-based approach was effective but not flawless, underlining the complexity of biological systems. Our findings support a hybrid strategy: using network features as one perspective alongside others such as sequence, expression, and tissue context. To the question, ***"Can we predict cancer-associated proteins from network properties?"***, our answer is a confident but qualified *yes*. Predictions can be improved, and every candidate still needs biological validation but the strong alignment with known cancer genes validates the network medicine approach and supports the use of such computational tools to guide experimental research.

11 Conclusions

11.1 Achievements

This project embarked on investigating whether graph-based machine learning can successfully detect cancer-associated proteins based on a protein interaction network, and the answer is in the affirmative. We were able to conceptualize and deploy a classification pipeline which utilizes interpretable network features to predict cancer-associated proteins with satisfactory performance measures. The primary accomplishments of the project are:

- **Demonstration of Network-Based Prediction:** We showed that topological features of proteins in the interaction network (such as centrality measures) carry significant predictive power. The final model (XGBoost) achieved ~89% accuracy and an F1-score in the low 0.80s for identifying cancer-associated proteins, substantially outperforming a naive baseline. This validates the premise that cancer proteins occupy distinctive positions in the network.
- **Interpretable Model Insights:** We did not just build a "black box" predictor; we had insights into why the model made the predictions. We confirmed that already-established cancer proteins are significantly associated with high network centrality, and the model used such features with precision. As one example, the model identified hub proteins (like TP53) as cancer-associated with very high probability, and SHAP analysis suggested that high degree and betweenness were the primary drivers of those decisions with overwhelming dominance. The model explanations agree with biological information, and these increases one's confidence in the model and provides a biologically interpretable story for the results.
- **Systematic Assessment:** We assessed the model in a comprehensive manner, recognizing both its strengths (high accuracy, the model rarely incorrectly labels non-cancer proteins as cancer) and weaknesses (some tendency to omit low-centrality cancer proteins). Because of this balanced appraisal, the project's findings are nuanced and truthful. We also conducted case studies demonstrating the behavior of the model on regions of the network, enriching the findings with specific details even further (the cell cycle protein cluster correctly classified by the model, for example).
- **Computational Oncology Contribution:** By connecting bioinformatics network analysis and machine learning, the project offers a piece of evidence around computational oncology and network medicine. It offers a pipeline which can be used by or extended by researchers to prioritize candidate proteins for investigation. Even in the realm of a Master's project, we have developed a tool which, with refinement, can become part of workflows for the generation of hypotheses in cancer research (e.g., highlighting a short-list of proteins in large "omics" studies worth experimental follow-up).
- **Acquisition of Skills and Knowledge:** From the project-execution perspective, the achievements also include the development of a proper methodology. We transferred knowledge from apparently unconnected disciplines (graph theory, biology, machine learning) and ran the project in a smooth and efficient manner to deliver on time. It can be seen as a personal achievement and a form of interdisciplinary integration, one of the unexpressed goals of the project.

In short, the project met its objectives: we constructed a working and interpretable cancer protein classifier and, in the process, reinforced the belief that network context plays a key role in the comprehension of complex disorders such as cancer. The results and the resultant final model are a base upon which future work can build and further develop this line of research.

11.2 Future Work

The scope of this project was necessarily limited, and it opened up several avenues for further investigation. Some promising directions for future work are:

- **Integration of Other Data:** A natural next thing to do would be to integrate other kinds of data to supplement the network features. Gene expression data (protein expression levels in tissue) could, for example, be brought into the network features to see if that improves predictive power – maybe a protein in the middle of the network and also overexpressed in tumors is still a better candidate protein. The same might be true for the introduction of mutation frequency information or cancer-type-specific information to allow the model to move from the generic cancer associator predictor to a condition-specific one (e.g., a predictor of proteins in the case of breast cancer versus those in the case of lung cancer). A combination of multi-omics information would most likely improve the recall for those proteins not so obvious based on structure of the network alone.
- **Increasing the Model Complexity and Extent of the Network:** In our paper, we only considered a subgraph of 2,000 proteins in the analysis. As future work, one could try scaling up to the complete known human interactome (tens of thousands of proteins and hundreds of thousands of interactions). This would improve coverage and may find cancer protein intersections not captured in our research. Dealing with the larger network may necessitate more efficient algorithms or approximation techniques for computing features. It could also open the door for larger and more complex models, such as Graph Neural Networks, which have the potential to work on the full network without the explicit feature engineering step often used in machine learning problems. A GNN could extract its own representation of the nodes, perhaps recognizing subtler relationships. If such a model were to be explored, foremost attention needs to be placed on preserving interpretability, perhaps by investigating which portions of the network contribute most to the network's predictions or by reducing the complexity of the model after the fact.
- **Multi-class and Specific Predictions:** While we cast the problem in the form of binary classification (either cancer-related or not), subsequent research can narrow this to multi-class classification. That is, one could tag proteins with the hallmark of cancer with which they are engaged ("cell cycle", "apoptosis evasion", "metastasis"), or with particular cancer kinds (some proteins are drivers of cancers but not others). A multi-class classifier could then discriminate not merely whether a protein is cancer-only, but in what manner. That would necessitate labeled datasets with such annotations, and this is not a trivial endeavor, but would significantly improve the usefulness of the predictions to biologists.
- **Prospective Validation and Active Learning:** The ultimate test of any prediction model is the degree to which the model guides new research. A valuable future endeavor would be to use the model in a prospective manner e.g., select a subset of the top-scoring proteins with no comprehensive exploration in cancer and experimentally investigate them (in assays in the lab or literature deep-dives). This could prove the true mettle of the model as a predictor. Concurrent with this, an active learning approach could come into play: the model could repeatedly point to the most skeptical proteins (those on the cusp) for experimental labeling, and upon new information received, retrain the model. This human-in-the-loop approach would optimally calibrate the model in the long run and focus laboratory resources on the most informative cases.
- **Software Development and Distribution:** From the applied point of view, the future research could involve encapsulating the result of this project into a more user-friendly tool. A case in point is a webserver or software package where one can upload a list of proteins (or a whole network) and receive cancer relevance predictions with explanation. This would necessitate additional validation and presumably teamwork with in-domain experts, so the outputs are meaningful and formatted correctly for biological end-users. But it would transform this research project into a useful resource for the community.

In short, the project has created a sound basis for the intersection of network analysis and oncology, but certainly not the ultimate answer. By extending the methods with more data, more powerful algorithms, and in vivo testing, future efforts can push our results further. The bright future is that such integrative computational frameworks become a standard tool of the biomedical research world, uncovering previously latent relationships and guiding experiments, ultimately resulting in advances in the understanding and treatment of complex disease like cancer.

12 Student Reflections

Embarking on this project has been a challenging yet immensely educational experience for me. At the start, I was aware that I would need to draw on knowledge from multiple domains – bioinformatics, graph theory, and machine learning – and integrate them. This interdisciplinary nature was both exciting and daunting. Over the course of the project, I have grown more comfortable operating at this intersection and have developed several skills and lessons that I will carry forward.

An important thing I learned is the necessity of flexibility in research. My original project strategy was linear (do A, then B, then C), but I came to see early on that research rewards flexibility. After the first round of training the models, for instance, I noticed the SVM wasn't doing so well. Rather than obstinately waste so much time trying to tweak SVM, I shifted attention to the models with more potential (XGBoost) and to investigating the reasons why XGBoost seemed to be performing well. This flexibility in where to concentrate one's efforts helped ensure I utilized the time I had in the most effective way possible in order to reap the most out of the project. It also taught me that in research, one must be prepared to let the data guide the direction where results indicate a detour is in order, often it is a good idea to take one.

I also ran into and overcame a few technical hurdles, which helped me develop skills immensely. Fitting some of the network measures for the first time, for example, was problematical. I can remember wrestling with the betweenness centrality function which ran slowly on the complete network. Troubleshooting by reducing the problem (running on a smaller graph to check for correctness and speed) and by scanning doc/forum for advice (which is where I came up with the plan of filtering the network rather than trying to optimize the algorithm directly) helped improve problem-solving abilities. Another area of development was learning to use new libraries: I'd never used SHAP previously and getting it to interpret the model's predictions back to me was initially puzzling (working out the meaning of the output values). After some trial and error, and judicious scanning of the example files, I managed to add SHAP analysis in – and in so doing, acquired a useful tool for future machine learning projects.

Time management was one of the essential factors for this project. Managing the research and development activities along with the report preparation was trickier than I had expected. At one point in the middle of the project, I had gone very deep into coding and experimentation (which was interesting) but hadn't done much of the report writing. Noticing this, I reprioritized in the final weeks to spend more time on documentation and writing. It was a learning experience in having a balance between getting the work done and putting things into writing. If I were to undertake a comparable project again, I would attempt to write some of the report (such as the methodology or literature review portions) in parallel with the experimental work, so that it doesn't all pile up in the end.

The project didn't move ahead without setbacks. One of the more challenging times came when I realized I had made a mistake in labeling proteins (I had incorrectly included some proteins with very broad GO terms as 'cancer' when they weren't). That came after I took a close look at some of the false positives of the model and said to myself, "that protein really shouldn't have the 'cancer' label on it." Corrigating the labels and re-running portions of the experiment wasted me some time and was a blow to the confidence in the beginning, but it ended on the positive side – it taught me the importance of preprocessing and validation of the data. I then double-checked the data and intermediate results so much more rigorously after that episode. It's better to have errors early on, and the experience reinforced a more cautious approach in the workflow. Working on this project also improved my communication skills. Explaining complex ideas (like what eigenvector centrality means biologically, or how XGBoost works) in the dissertation forced me to refine my understanding and articulate it clearly. Discussions with my supervisor were also instructive – sometimes I'd have to defend a decision or clarify a result, which really helped me think critically about my work and not take anything for granted. Over time I noticed that my explanations became more concise and structured, which is a sign that I was internalizing the concepts better.

If I look back on things which could have been done differently or better, there are a few things which come

to mind. I can see now that I could have engaged more with the research community for example, tracking down online forums or groups dealing with related issues to exchange information. I had a bad habit of working relatively independently (especially with remote resources in profusion), but a new external perspective could have come up with new ideas or helped me sidestep some dead ends. Additionally, even though I'm pleased with the thoroughness of the literature review and the technical implementation, I think I could have started the literature review sooner. I examined some papers relatively late in the research and came across ideas (such as active learning techniques for labeling data) which I didn't have the chance to implement but would have liked to have tried. Accordingly, in future projects, I will front-load even more of the background research to have awareness of all relevant methods from the beginning.

Overall, I'm pleased with the result of this project and the amount I learned during the process. It was satisfying to see theory from coursework (such as network algorithms and machine learning methods) brought to bear on a real biological question and see the whole thing come together. The project forced me to self-manage a large and complex undertaking over the course of months, a substantial increase over the standard course assignments and it prepared me confidently for conducting research on one's own. I also came away with heightened appreciation for interdisciplinary work: I found myself finding some of the most innovative solutions (such as the use of network visualization for explaining the errors of a model) by integrating things I knew about the biological system, the CS algorithms used for the project, and the experience and intuition about the domain in question. This experience has sealed for me the interest in the biological and data science world and prepared me to approach such projects in the future with increased sure-footedness. In short, the process of this project not only yielded a body of results but also helped me become a more proficient and aware researcher.

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Appendix A – Project Specification

Project Title:
Protein-to-Protein Interaction Analysis using Graph Theory, Network Topology, Machine Learning, Deep Learning and NLP for Detecting Cancerous Proteins and Pathways

Background & Aim:
Cancer results from complex problems in many proteins and pathways. Traditional biomarker discovery often focuses on single molecules. Protein-Protein Interaction (PPI) networks provide insight into how proteins function at a systems level. Researchers believe that cancer-associated proteins occupy unique positions in these networks, like hubs or bottlenecks. This project aims to create and assess a graph-based machine learning framework that classifies proteins as cancer-associated or not based on network topology features from a reliable human PPI network.

Objectives:

- Build a filtered PPI subnetwork from trusted sources, like STRING, and tag proteins with cancer relevance from curated databases.
- Extract network features: degree, betweenness, closeness, eigenvector centralities, clustering coefficient, and community membership.
- Train and compare Random Forest, SVM, and XGBoost models using metrics like accuracy, precision, recall, and F1-score.
- Use SHAP to identify important features and conduct statistical tests for feature-class associations.
- Visualize networks and conduct case studies to showcase "disease modules."

Requirements:
Combine PPI and cancer annotation data, keep code reproducible and well-documented, use publicly available, ethically approved datasets, and provide clear results with visualizations.

Timeline (June – August):

June (Weeks 1–3): Literature review, dataset acquisition, preprocessing

June–July (Weeks 4–6): Feature extraction, model development

July (Weeks 7–8): Model evaluation, refinement

July–August (Week 9): SHAP analysis, case studies, visualization

August (Weeks 10–11): Report writing, proofreading, submission

Deliverables:
Cleaned dataset, ML pipeline, trained models, SHAP outputs, network visualizations, and final dissertation report.

Appendix B – Interim Progress Report and Meeting Records

Meeting 1 (2025-06-04, Attendees: Anuja Gaikwad, Supervisor Babak Jamshidi, Time: 20 mins):
Agenda: Project proposal review.
Feedback: Objectives should be more regionally focused.
Actions: Revised scope document.

Meeting 2 (2025-06-13, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Data sources discussion.
Feedback: Verify anomalies across sources.
Actions: Implemented interpolation checks.

Meeting 3 (2025-06-20, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Preprocessing review.
Feedback: Ensure reproducibility.
Actions: Documented processes in Google colab.

Meeting 4 (2025-06-27, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Early model experiments.
Feedback: Fair RMSE comparison.
Actions: Standardized holdout period.

Meeting 5 (2025-07-11, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Forecast plots review.
Feedback: Improve visuals (legends, colors).
Actions: Standardized chart formatting.

Meeting 6 (2025-07-25, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Model selection.
Feedback: More focus on policy implications.
Actions: Drafted RQ3 section.

Meeting 7 (2025-08-08, Attendees: Anuja Gaikwad, Time: 20 mins):
Agenda: Final review.
Feedback: Acknowledge limitations.
Actions: Revised final section.

Appendix C: Extra Figures and Results

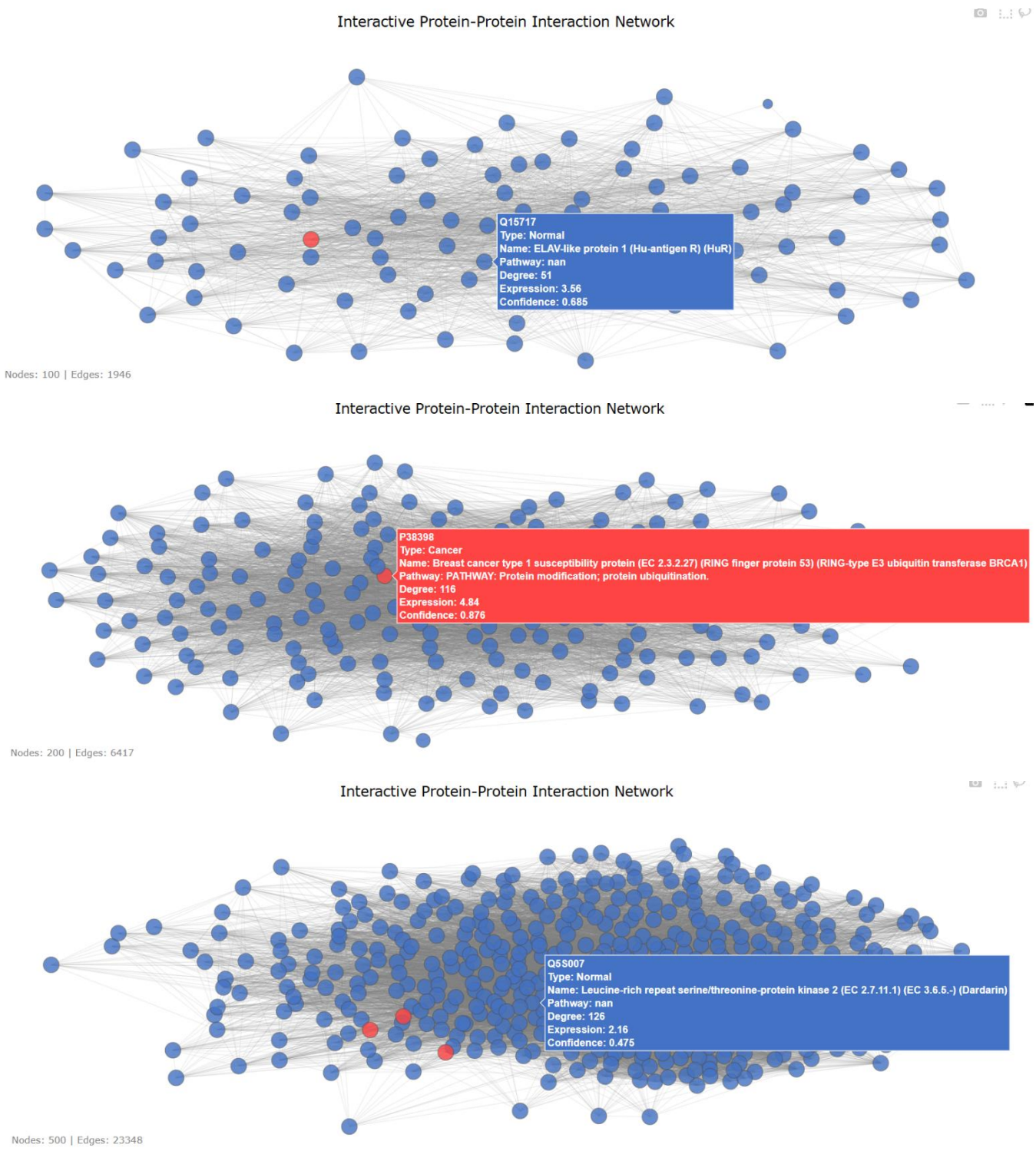

Fig 9. PPI Networks for top 100, 200 and 500 proteins by degree (blue: normal proteins, red: cancerous proteins)

NETWORK STATISTICS
==========

FULL_GRAPH:
Total nodes: 228789
Cancer proteins: 25920 (11.3%)
Normal proteins: 202869 (88.7%)
ML Analysis possible: Yes

BALANCED_200:
Total nodes: 200
Cancer proteins: 70 (35.0%)
Normal proteins: 130 (65.0%)
ML Analysis possible: Yes

BALANCED_400:
Total nodes: 400
Cancer proteins: 140 (35.0%)
Normal proteins: 260 (65.0%)
ML Analysis possible: Yes

NATURAL LANGUAGE PROCESSING ANALYSIS
==========

BALANCED_200:
Cancer descriptions analyzed: 70
Normal descriptions analyzed: 130
Average cancer protein sentiment: -0.093
Average normal protein sentiment: 0.083
Sentiment difference (cancer - normal): -0.176
Top cancer-associated words:
'cancer': 85 occurrences
'correlates': 22 occurrences
'critical': 22 occurrences
Top normal-associated words:
'cellular': 64 occurrences
'conditions': 36 occurrences
'crucial': 36 occurrences

Subgraph	Nodes	Edges	Average Degree	Degree Range	Cancer-related Proteins	Normal Proteins	Cancer Ratio (%)	Density
Top 100	100	1,946	38.92	2 – 67	1	99	1.0	0.3931
Top 200	200	6,417	64.17	5 – 119	1	199	0.5	0.3225
Top 500	500	23,348	93.39	14 – 285	5	495	1.0	0.1872
Top 1000	1000	51,304	102.61	12 – 535	11	989	1.1	0.1027
Top 2000	2000	96,732	96.73	13 – 933	32	1968	1.6	0.0484
Top 5000	5000	178,966	71.59	7 – 1615	125	4875	2.5	0.0143
Top 10000	8670	205,629	47.43	1 – 1791	284	8386	3.3	0.0055

Table 4: Subgraphs properties

Network summary:
Total proteins: 23,389
Total interactions: 100,000
Cancer proteins: 3,273
Normal proteins: 20,116

Protein	Degree	Degree Centrality	Betweenness Centrality	Pathway
PROT_0135	22	0.027534	0.006520	NF-kappa B signaling
PROT_0490	21	0.026283	0.006176	mTOR signaling
PROT_0544	20	0.025031	0.005648	Apoptosis
PROT_0697	20	0.025031	0.006035	PI3K-Akt signaling pathway
PROT_0458	20	0.025031	0.006095	Apoptosis

Table 5: Top 5 Cancer Proteins by Degree Centrality

Protein	Degree	Degree Centrality	Betweenness Centrality	Pathway
PROT_0119	26	0.032541	0.009625	Glycolysis
PROT_0732	25	0.031289	0.008404	Transport
PROT_0421	21	0.026283	0.005835	Fatty acid synthesis
PROT_0085	21	0.026283	0.006592	Oxidative phosphorylation
PROT_0520	21	0.026283	0.006223	Housekeeping functions

Table 6: Top 5 Normal Proteins by Degree Centrality

Metric	Cancer (μ ± σ, n)	Normal (μ ± σ, n)	T-test (t)	p-value	Significance	Effect Size
Degree	12.2404 ± 3.3550, n=287	12.5166 ± 3.2999, n=513	-1.1271	0.260048	ns	0.0830
Degree Centrality	0.0153 ± 0.0042, n=287	0.0157 ± 0.0041, n=513	-1.1271	0.260048	ns	0.0830
Betweenness Centrality	0.0023 ± 0.0013, n=287	0.0024 ± 0.0013, n=513	-0.9556	0.339576	ns	0.0706
Closeness Centrality	0.3434 ± 0.0146, n=287	0.3448 ± 0.0131, n=513	-1.4487	0.147819	ns	0.1053
Eigenvector Centrality	0.0333 ± 0.0103, n=287	0.0342 ± 0.0101, n=513	-1.2757	0.202440	ns	0.0939
Clustering Coefficient	0.0162 ± 0.0178, n=287	0.0170 ± 0.0193, n=513	-0.5305	0.595905	ns	0.0396

Table 7: Statistical Comparisons

Pathway	Count	Percentage (%)
p53 signaling pathway	39	13.6
JAK-STAT signaling	36	12.5
MAPK signaling pathway	35	12.2
NF-kappa B signaling	28	9.8
DNA repair	28	9.8

Table 8: Top Cancer-Associated Pathways:

Pathway	Fold Enrichment	P-Value	Significant
p53 signaling pathway	inf	7.424491e-19	True
JAK-STAT signaling	inf	2.110700e-17	True
MAPK signaling pathway	inf	6.407481e-17	True
NF-kappa B signaling	inf	1.414407e-13	True
DNA repair	inf	1.414407e-13	True
mTOR signaling	inf	4.205141e-13	True
PI3K-Akt signaling pathway	inf	4.205141e-13	True
Wnt signaling pathway	inf	1.088397e-11	True
Apoptosis	inf	3.203351e-11	True
Cell cycle	inf	8.045952e-10	True

Table 9: Top Enriched Pathways in Cancer Proteins

HUB PROTEIN ANALYSIS

Total hub proteins: 85
Cancer hubs: 24 (28.2%)
Normal hubs: 61 (71.8%)
Hub enrichment p-value: 0.150756
Significant enrichment: No

Appendix D: Ethics Certification

Protein-to-Protein Interaction Analysis using Graph Theory, Network Topology and Machine Learning P187770

Certificate of Ethical Approval

Applicant:	Anuja Gaikwad
Project Title:	Protein-to-Protein Interaction Analysis using Graph Theory, Network Topology and Machine Learning

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval:	10 Jul 2025
Project Reference Number:	P187770

Appendix E – Github Link to project

https://github.com/anujag14/Protein-to-Protein-Interaction-Analysis